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Impact and characteristics of nutritional programs in improving the nutritional status of adult people living with HIV in sub-Saharan Africa: A systematic review of evidence

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-047205
Article Type:	Original research
Date Submitted by the Author:	23-Nov-2020
Complete List of Authors:	Tesfay, Fisaha; Mekelle University College of Health Sciences, School of Public Health Javanparast, Sara; Flinders University, College of Medicine and Public Health Gesesew, Hailay ; Flinders University, College of Medicine and Public Health Mwanri, Lillian; Flinders University, College of Medicine and Public Health Ziersch, Anna; Flinders University, Southgate Institute for Health, Society and Equity
Keywords:	PUBLIC HEALTH, NUTRITION & DIETETICS, EPIDEMIOLOGY

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Impact and characteristics of nutritional programs in improving the nutritional status of adult people living with HIV in sub-Saharan Africa: A systematic review of evidence

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Abstract

Objectives: Although some studies have identified various challenges to the effectiveness of nutritional programs, existing evidence varies by country context, program goals, and implementation.This systematic review aims to synthesise the existing evidence on the impact of nutritional programs on nutritional outcomes of people living with HIV in sub-Saharan Africa.

Design : Systematic review

Data sources : We searched for primary studies published in various databases, google, google scholar and reference of refences.

Eligibility criteria: Studies published since 2005 and reported on the nutritional oucomes were included.

Data extraction and synthesis: Data was extracted using data extraction perfoma and nutritional outcomes of people living with HIV before and after enrolment in a nutritional program were compared and analysed using thematic synthesis.

Results: Sixteen studies assessing the effectiveness of nutritional programs in HIV care were included**.** Of these, 13 examined nutritional programs that were operational in health facilities while the remaining was given outside of the health facility. Nutritional recovery (defined differently in the studies) ranged from 13.1% to 67.9%. Overall program failure rate, which includes default after enrolment in a nutritional program or non-response ranged from 37.6% to 48%. More specifically, non-response to the nutritional program ranged from 21% to 67.4% and default from the program ranged from 19% to 70.6%. Key sociodemographic, clinical, and nutritional characteristics that affect, nutritional recovery, non-response, and default were also identified.

Conclusions and recommendations: Nutritional programs in HIV care have led to improvements in nutritional outcomes. However, the programs were characterised by a high magnitude of default and non-response. To improve favorable nutritional outcomes of people living with HIV, a holistic approach that addresses longer-term determinants of undernutrition is needed.

Keywords: Nutritional program, nutritional support, HIV and nutrition

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Strength and limitations

- This is the first systematic review on nutritional outcomes of people living with HIV in Sub Saharan Africa
- The review assesses multiple nutritional outcomes and presents summary evidence of a range of nutritional programs
- The diversity and heterogeneity of the studies and nutritional programs meant that meta-analysis was not able to be done.
- The review brings together and consolidates findings from a range of studies, including a number that had small samples, providing a consolidated overview of evidence of nutritional outcomes for nutritional programs in HIV settings, and predictors of effectiveness.

The original protocol for the study: The protocol of the study is attached at the end of the study.

Funding statement: Not funded

A competing interests statement. All authors in this study declare that they have no conflict of interest.

Introduction

Existing evidence demonstrates the clinical and social susceptibility of people living with HIV to undernutrition and its wider effects on individuals, families, and communities [1 2]. Inadequate food intake due to lack of appetite, malabsorption, and impaired storage and metabolism are clinical causes of undernutrition among people living with HIV [3] while the social factors include poverty, gender inequality, food insecurity, stigma, and discrimination [4 5].

Undernutrition has various impacts on the wellbeing of people living with HIV including HIV service utilisation. The impact of undernutrition on HIV-related mortality in poor resource settings has been highlighted in many studies [6-8]. Research has also indicated that undernutrition can act as a barrier to the utilisation of HIV care services and antiretroviral therapy (ART) adherence [9-11].

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The World Health Organisation (WHO) identifies nutritional intervention as a critical component of HIV management and recommended the integration of nutrition into HIV care and support services [12] [13]. To address the impact of undernutrition among people living with HIV, nutritional programs targeting people living with HIV are operational in resource-poor settings, mainly in sub-Saharan Africa [14]. The key defining characteristics of the nutritional programs in HIV care in these settings is nutritional assessment, counseling, and support services [15-17]. The nutritional support component of the nutritional program includes the provision of micronutrient, macronutrient, or fortified macronutrient supplementation [18] [19].

Nearly 90% of HIV care sites in sub-Saharan Africa have responded to WHO recommendations and instituted some form of nutritional program [20], with substantial variations between and within countries [16] [18] [21]. Various factors including the policies of donor organisations regarding the nutritional program and availability of national and local resources to implement the programs have contributed to the differences observed between countries in sub-Saharan Africa [22], including the type of care [23-25] and nature of nutritional support provided [26-28]. However, the existing evidence regarding the impact of nutritional programs on nutritional outcomes is a fragment, mixed, and varies by country context, program goals, and implementation. This review aims to synthesise the existing evidence on the impact of nutritional programs on nutritional outcomes of people living with enrolled in nutritional countries in HIV care.

Methods

Protocol

The protocol for this systematic review has been registered under the Prospero review registration number CRD4202019682.

Systematic search strategy

This review was guided by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). Studies published in English language Web of Science, Medline, Scopus, ScienceDirect, ProQuest, from 2005 to the 10th of July 2020 were searched. Google Scholar, google search, and the reference lists of references of selected papers were also searched. The search concepts and terms included “Nutritional programs”, “HIV/AIDS”, and “sub-Saharan Africa” (**table 1**).Outputs of Ovid medline is given in the supplemetal information (**supplementary file 1**).

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Table 1: Concepts and their respective keywords

Concept 1: ***Nutritional program***	**Concept 2:** ***HIV/AIDS***	**Concept 3:** ***Sub-Saharan Africa***
Nutritional program	HIV-1	South of Saharan Africa
Nutritional intervention	HIV-2	Africa
Nutritional care	HIV infection	Western Africa
Food by prescription	Human immunodeficiency syndrome	Central Africa
Supplementary food	Acquired immune deficiency syndrome	
Ready to use Therapeutic food		
Food supplementation	AIDS	List of all countries in sub-Saharan Africa

Selection of studies

The screening process involved two stages. The first stage was conducted by FT and involved screening the studies by title and abstracts against the inclusion criteria. The second stage was conducted by two independent reviewers (FT and HAG) and involved full-text screening. The inclusion criteria were:

- longitudinal studies (cohort and interventional) that involved adults (age>15 years)
- reported enrolment into a nutritional program
- measured nutritional outcomes using BMI, weight gain or MUAC
- published in English since 2005, because most nutritional programs in sub-Saharan Africa started in 2005.
- primary studies on nutritional outcomes of people living with HIV who were enrolled in nutritional programs.

Studies were excluded from the review if information about age or pregnancy status was not provided because different enrolment and nutritional outcome criteria are used to enroll children and pregnant women in the program (**Figure 1**).

Measurement of exposures

The exposure variables included: sociodemographic factors such as age, sex, residence, marital status, and educational status; clinical, such as WHO clinical stage, CD4 count, functional status, and presence of opportunistic infections, HIV status disclosure; and

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nutritional factors such as baseline nutritional status, the status of food security and other dietary information.

Measurement of outcomes

The outcome variables in the review were nutritional outcomes of adults living with HIV, compared before and after enrolment in a nutritional program. The nutritional outcomes included change in BMI, weight, or MUAC and nutritional recovery (BMI≥ 18.5 kg/m^2), default, or non-response to the nutritional program.

Context/setting

Nutritional programs in sub-Saharan Africa commenced at various times depending on the country context. Most nutritional programs in sub-Saharan Africa have similar program components which include nutritional assessment, counseling, and nutritional support. Because of the context, duration, enrolment, and exit criteria, nutritional programs are different from other settings that are why the current review is limited to studies conducted in sub-Saharan Africa only.

Assessment of methodological quality

The Joanna Briggs Institute [29] for critical appraisal tool for quasi-experimental studies was used to assess the quality of studies that were included in the review. Two independent reviewers (FT and HAG) conducted the quality assessment.

Data extraction

Data were extracted using a data extraction proforma adopted from JBI-MAStARI. The extracted data involved the outcome and exposure information that may affect the nutritional outcomes of people living with HIV/AIDS. In addition, population, sample size, country, sample size, description of the nutritional program, and the conclusion of the studies were extracted.

Data analysis

Due to the heterogeneity of the studies included in terms of study design, the measure of nutritional outcomes, and differences in enrolment and exit criteria, a narrative synthesis was the most appropriate analysis procedure for this review. Different variables were identified from the reported findings of each study and those with similar patterns were grouped. In subsequent coding and analysis, categories and themes were generated. Hence, the characteristics of nutritional programs in sub-Saharan Africa and the impact of the nutritional program on nutritional outcomes are the two major themes.

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Patient and Public Involvement: No patient involved. This is a systematic review and it is nor applicable to make patient and public involvement.

Results

Description of the studies

A total of 2006 and 30 papers respectively were obtained from electronic databases and through other sources. After removing duplicates 1,738 studies were screened by title and abstract and 1,708 were irrelevant. 30 papers were selected to be scrutinised further against the inclusion criteria using full-text screening, 16 papers were included in the final review and 14 studies were excluded (**Image 1**). Of the 16 studies included in the final review involving a total of 25,335 participants, nine were retrospective cohort, three were prospective cohort, and four were randomised trial study designs.

Assessment of methodological quality

The methodological and reporting quality of the reviewed studies was assessed using the JBI critical appraisal tool for quasi-experimental studies (**supplementary file 2**). The assessment tool included nine parameters each carrying equivalent weight (**supplementary file 3**).

Nine studies, six clinical trials [30-35], and three cohort studies [25] [36] [37] scored nine out of nine (100%).

One retrospective cohort study [38] scored seven out of nine but the two parameters related to the comparability of study participants and similarity of interventions in both the exposed and non-exposed groups were not applicable− scoring 100%. Three cohort studies scored six out of nine which is 100% as two parameters such as comparability of participants and similarity of interventions were not applicable and the presence of control groups was not clear [39-41].

Two retrospective studies scored eight out of nine (89%) as there was no control group [42] [43]. Another retrospective cohort study scored six out of nine (86%) where the comparability of study participants and intervention was not applicable, no control group, and unclear [44].

Characteristics of the nutritional programs

The nutritional programs examined in this review varied in terms of enrolment criteria, duration, and type of nutritional support used. Six studies used BMI <18.5 kg/m^2 as an enrolment criteria [25] [33] [34] [40] [42] [44]. Another seven studies used one of these− bilateral oedema, being bedridden for 50% of the month, TB coinfection, CD4 count < 350, WHO clinical stage III and IV, in addition to BMI< 18.5 kg/m^2 or MUAC <210mm) [31] [32] [37-39] [41] [43]. Living in a household with HIV positive family members was used as an enrolment criterion in one study [36]. Another study enrolled people living with HIV at any stage of the disease and not able to eat [30]. A further study used

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the world food program (WFP) poverty assessment index, no receipt of other food assistance, CD4 count between 200 and 450 cells per microliter, and ART naïve (new for the treatment) as an enrolment criterion [35].

In five studies, the duration of the nutritional program ranged from three (mildly and moderately) to six months (severely) undernourished [25 38 39 42 44]. Two studies reported nine weeks duration [33 44], another two 14 weeks [32 34], and a further study 24 weeks [31] duration before individual patients exited from the program. The nutritional program was nine months in one study [37] and twelve months in three studies [33 35 36]. No duration limit was reported in two studies [40 43].

Various types of nutritional supports were used in nutritional programs. Ready to use therapeutic food (RUTF) or ready to use supplementary food (RUSF) were used in eight studies [25 30 31 33 38 39 42-44]. Corn/soy blended food was used in three studies [32 34 37 40] and the WFP food basket (which involves generic food rations) was used in two studies [35 36]. One study used an extract of hypoxis, grapefruit seed extract, sitosterol and sitosterolin, beta-carotene, vitamin E, vitamin, vitamin B1, vitamin B2, vitamin B12, nicotinamide, vitamin C, olive green leaf extract, folic acid, and natural antioxidant (biocydin) as the nutritional supplement [41].

The nutritional programs were provided in health facilities such as hospitals and clinics in 12 studies [25 30-34 36-38 42-44], in community-based settings in four studies [33 39 40], and no setting was specified in one study [41].

Impact of the nutritional program on nutritional outcomes

In reporting improvements in the nutritional outcome of people living with HIV, studies used various outcome measures. Gain in BMI, weight, and MUAC were used either in combination or as separate nutritional outcome measures. BMI only was used in nine studies [25 34 35 37 38 41-44] and both MUAC and BMI were used in two studies [31 39]. The other two studies used weight gain [30 36], and three studies used both BMI and weight gain to measure the nutritional outcomes [30 33 36].

Evidence from most studies showed that nutritional outcomes included nutritional program failure, recovery, default, and non-response. The definition of these terms in the studies included are shown in **table 2.**

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Table 2: Shows the definition of nutritional outcomes in nutritional programs

Outcomes	Adults
Program failure	Includes default and non-response from the nutritional program
Improvement in nutritional status and recovery /graduation from the nutritional program	Various criteria were used to determine nutritional recovery • Nutritional Reaching BMI of 18.5kg/m^2 for two consecutive visits in some studies or BMI>18 kg/m^2 or BMI>20kg/m^2 In other studies, the nutritional outcome was measured in terms of improving weight, BMI, or MUAC.
Non-response	Participants did not reach a BMI of 18.5kg/m^2 for two consecutive visits or did not show any change in BMI, weight, and MUAC
Default /failure to complete the nutritional program	Dropped out of the program before reaching the target BMI, weight or MUAC

Improvement in nutritional status or Recovery from the nutritional program

Eleven studies measured nutritional recovery, defined differently in the various nutritional programs. Three studies defined program recovery as BMI>18.5kg/m^2 in two consecutive measurements [25 42 44], and nutritional recovery in these studies ranged from 32.6% [25] to 62.2% [44]. One study used BMI >18 kg/m^2 in two subsequent measurements and the absence of oedema, and recovery was 47.4% [38]. A further study defined program recovery as BMI>20kg/m^2 and 13.1% attained nutritional recovery against these criteria [40].

Four studies used improvements in BMI, without specifying a specific cut-off point [34 35 37 43] and the improvement in BMI in these studies ranged from 0.6 kg/m^2 using WFP rations for 12 months [35] to 2.2 kg/m^2 in a study where participants received a fortified spread for 14 weeks

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[34]. Those who used a corn-soy blend showed a BMI increase of 1.7 kg/m^2 [34]. Weight gain was used to measure nutritional outcomes in four studies [30] [33] [36] [39] and weight gain ranged from 0.85 kg [33] to 3 kg [39].

One study using weight gain found that after 9 weeks of supplementation, body weight, and fat-free mass increased significantly by 11 % (p= 0.033), and 11.8 % (p= 0.033) in the RUTF group, but not in the control group [30].

One study used BMI, weight gain, and MUAC [31], and those in the RUSF group had significantly greater gains in weight, BMI, and MUAC than those in the no-RUSF group (p=0·004, and 0·03, respectively).

Table 3 presents the socioeconomic, demographic, clinical, and nutritional determinants of nutritional outcomes among those enrolled in nutritional program. For sociodemographic factors that affected nutritional outcomes, female gender was associated with nutritional recovery in three studies [44] [25] [40]. Compared with those who didn't attend any schooling, individuals who attended school were more likely to recover than those who didn't in one study [44], and attending the nutritional program in a hospital was inversely related to nutritional recovery in one other study [25].

Few clinical characteristics were associated with nutritional recovery. In one study, not being on ART was associated with nutritional recovery [40]. In another, compared with the WHO clinical stage 4, being at WHO clinical stage 1 and 2 were associated with nutritional recovery [44]. In addition, the absence of opportunistic infection at enrolment to the nutritional program was associated with nutritional recovery compared with those who had an opportunistic infection [44].

Regarding the impact of nutritional characteristics at enrolment, compared with those enrolled with severe undernutrition, those with moderate undernutrition to the nutritional program were associated with nutritional recovery in two studies [40] [44]. In one study, enrolment in the nutritional program with mild undernutrition was associated with nutritional recovery compared with those with severe undernutrition [44]. In another study, those who received the nutritional program were more likely to cover, compared with those who did not receive the nutritional program [25].

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Table 3: Improvements in nutritional status and recovery and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food type	Improvement in nutritional status or recovery	Factors that affect nutritional recovery
Berhe, et al 2014 [44], Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUTF	328 (62.2%) recovered **Measurement**: BMI>=18.5 kg/m^2 in two subsequent measurements	The factors that were associated with nutritional recovery were: - female (adjusted odds ratio (AOR) and 95% confidence interval (CI)= 2.16 (1.340-3.473)) - educated (AOR and 95%= 1.82(1.110-2.983)) - moderate malnutrition (AOR and 95%= 2.594 (1.251-5.377)) - mild malnutrition (AOR and 95% =14.25 (7.074-28.742)) - WHO stage I and II (AOR and 95% CI =3.345 (1.652-6.772)) - no opportunistic infection during admission (AOR and 95% =2.791(1.047-7.439))
Mallewa, et al 2018 [31], Africa	Randomized clinical trial	-Adults and Adolescents living with HIV -At least five years with HIV infection and CD4 cell count fewer than -100 -Starting ART at the health facilities	1805 -897 to treatment -908 to non RUSF	Peri-urban regional hospital **Food type**: RUTF and RUSF	Adults and adolescents aged 13 years and older in the RUSF group had significantly greater gains in BMI, and MUAC than the no-RUSF group (p=0·004, and 0·03, respectively). **Measurement:** Improvement in weight, BMI and MUAC (no cut off point)	Not assessed

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Nagata et al 2014 [40], Kenya	Retrospective cohort study	-non pregnant adult patients -BMI less than 18.5 kg/m^2	1017	Rural village **Food type**: fortified corn-soy flour supplements	On average, enrolled clients gained 2.01 kg in weight and 0.73 kg/m^2 in BMI over follow-up (mean 100 days) with the highest gain among those severely malnourished. - Only 13.1% nutritional recovery according to criteria **Measurement**: BMI.20 kg/m^2	The factors associated with BMI >20 Kg/m^2 were - Greater BMI at baseline 1.52 (1.23–1.86) - younger age, 0.97 (0.94–1.00) - male gender, (0.51 (0.30–0.86)) - not requiring highly active antiretroviral therapy (HAART) (4.18 (2.42–7.22)
Ndekha et al 2009 [34], Malawi	Randomised control trial	- Adults living with HIV - BMI<18.5 Kg/m^2	491	Clinic based **Food type**: fortified corn-soy flour supplements	After 14 weeks, patients receiving fortified spread had a greater increase in BMI and fat-free body mass than those receiving corn-soy blend: 2.2 (SD 1.9) v 1.7 (SD 1.6). Measurement: Gain in BMI (no cut off point)	No significant differences in the CD4 count, HIV viral load, assessment of quality of life, or adherence to antiretroviral therapy were noted between the two groups
Oguntibeju et al 2007 [41], South Africa	Retrospective cohort study	- non pregnant adults with HIV, non-diabetic and non-hypertensive - not on ART - CD4 cell count	35	Not specified **Food type**: extract of hypoxis, grape fruit seed extract, sitosterol and sitosterolin, beta-carotene,	19 (67.9%) of the 28 patients had a BMI within the normal range after nutrient intervention **Measurement:** BMI>=18.5kg/m^2	- significant positive correlation between the BMI and fat percentage (statistics not reported) - CD4+T cell count showed no correlation with any of the anthropometric indices - the viral load significant negative correlation with the lean body mass and BMI (statistics not reported)

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		100-350 cells/mm^3		vitamin E, vitamin, vitamin B1, vitamin B2, vitamin B12, nicotinamide, vitamin C, olive green leaf extract, folic acid and natural antioxidant (biocydin)		
Olsen et al 2014 [33], Ethiopia	Randomised control trial	- Age greater than 18 years - BMI greater than 16 Kg/m^2	453	Referral hospital **Food type**: Plump sup	At three months, participants receiving the supplements containing whey or soy had increased their lean body mass by 0.85 kg (95% confidence interval 0.16 kg to 1.53 kg) and 0.97 kg (0.29 kg to 1.64 kg), respectively, more than controls. **Measurement**: Weight gain and BMI (no cut off point)	- No effect on physical activity - The effects of the two supplements, however, were not significantly different in direct comparison
Rawat et al, 2014 [35], Uganda	Prospective cohort	- household composition - ownership of valuable assets - employment	904	Community based **Food type**: WFP Rations	Over 12 months, food assistance significantly increased body mass index by 0.6 kg/m2 (P, 0.01) and mid-upper arm circumference by 6.7 mm (P, 0.05).	- no impact on CD4 count, Hb concentrations, or Individual Dietary Diversity Score - at the household level, food assistance increased the Household Food Insecurity Access Scale, by 2.1 points (P, 0.01)

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		status and income - housing characteristics and past experience of food insecurity			**Measurement:** Change in BMI (no cut off points)	- food assistance increased BMI by 0.6 kg/m2 and MUAC by 6.7 mm. - no statistically significant impact of food assistance on Hb concentration or CD4 count
Sadler, et al, [25] Ethiopia	Comparative follow up study. Report, not peer reviewed	Adults living with HIV - BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 interventio n were included in the final analysis	Health facilities / health centres **Food type**: RUTF/RUSF	636 (32.6) graduated in the intervention group and 120 (18.8) in the comparison group **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Factors that were associated with nutritional recovery were: - being in the intervention group 2.4 (1.6-3.4) - female 1.4 (1.0-1.9) - attending in a hospital 0.5 (0.4-0.7)
Scarcella. et al [43], Mozambique	Randomised control trial	- Adult HIV patients - BMI <18.5 Kg/m^2 - presence of wasting - TB co-infection - insufficient access to food - CD4 count < 200	106	Public Health Centre **Food type**: - food package consisted of a ration of cereals (rice or corn) of around 15kg; pulses (5kg); peanuts (3kg); sugar (2kg); sunflower seed oil (3Lt.); at least 7kg of fortified corn soy blend	In the cohort not in program the mean BMI values showed an increase but not significant (initial value: 21.9±2.9; final value: 22.5±3.3) **Measurement**: Increase in BMI (no cut off point)	- haemoglobin showed an increase but not significant - in the intervention group both, BMI and Haemoglobin shown a significant increase - increase in BMI was statistically associated with the dietary diversity score in the intervention group

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Tesfay et al, [42] Ethiopia	Retrospective cohort study	- Adults living with HIV - BMI<18.5	1757	Hospitals **Food type**: RUTF/RUSF	55.3% (95% CI = 53.2–57.4) recovered from malnutrition **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Not assessed
van Oosterhout et al 2010 [37], Malawi	Retrospective cohort study	-Adults aged 18 years and older -BMI 18.5 kg/m^2 -ART naïve or - WHO clinical stage III or IV Or - CD4 count B250 cells/ml irrespective of clinical stage	593	Centeral hospital **Food type**: corn/soy blended flour (CSB) or ready-to-use fortified spread (RUFS), a peanut butter-based product	- Linear regression analysis showed that increase in BMI was greatest in the RUSF group and better in the CSB group than in those receiving no food supplementation at 14 weeks. - These differences were no longer significant at 26 weeks **Measurement:** Increase in BMI	- no longer a significant effect of supplementary feeding on an increase in BMI, 12 weeks after the feeding was stopped. - factors that were associated with nutritional recovery or improved BMI was not assessed.

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Program failure rate

The program failure rate involves a combination of default, non-response, and mortality while enrolled in the nutritional program. The program failure rate was reported in two studies [38] [44]. The nutritional program failure rate in a study from Kenya and Uganda was 48% [38] and 37.6% in a study from Ethiopia [44]. Factors that were associated with the program failure were only assessed in one study [38] and found male gender, severe undernutrition at program enrolment (pre-ART), and ineligibility for ART at enrolment were associated with program failure (**Table 4**).

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Table 4: Shows program failure (default + nonresponse +death) after enrolment in a nutritional program in sub Saharan Africa

Author,year and country	Study design	Study population	N	Setting and type of food	Nutritional program failure	Factors that affect nutritional program failure
Ahou et al 2011 [38], Kenya and uganda	Retrospective cohort study	Adults aged 15 years and older living with HIV BMI less than 17 kg/m^2 and bilateral pitting oedema	1106	Regional hospital Type of food: RUTF/RUSF	An overall total of 531 of (48.0%) patients failed nutrition therapy	Factors that were associated with program failure were: - male gender (OR = 1.5, 95% CI 1.2-2.0), - patients with severe malnutrition at nutrition programme enrolment (OR = 2.2, 95% CI 1.7-2.8), and - those never started on ART (OR = 4.5, 95% CI 2.7-7.7) for those eligible; OR = 1.6, 95% CI 1.0-2.5 for that those ineligible for ART at enrolment
Berhe, et al 2014 [44], Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital Type of food: RUTF/RUSF	Overall program failure rate was 196 (37.6%)	The factors that were associated with nutritional program failure were not assessed.

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Non-response from the nutritional program

Three studies from Ethiopia measured non-response to the nutritional program and found a wide variation in rates. These rates were 21% [42], 29.6% [44], and 67.4% [25].

Compared with those who attended in a lower health facility (primary health care), attending a nutritional program in the hospital was associated with greater non-response in one study [25]. A study by Tesfay and colleagues examining hospital nutritional programs in Ethiopia reported that being an urban area was associated with non-response [42].

In terms of clinical characteristics, the same study found advanced WHO clinical stage (stage IV), and bedridden functional status were associated with non-response to the nutritional program [42]. Compared with those enrolled with mild undernutrition, severe undernutrition at enrolment was also associated with non-response [42].

In another study, those with severe food insecurity were less likely to not respond than those who were food secure [25]. The third study did not assess factors contributing to non-response (**Table 5**).

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Table 5: Nonresponse to the nutritional program and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food	Nonresponse to the nutritional program	Factors that contribute to non-response to the nutritional program
Berhe, et al 2014 [44], Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUFT/RUSF	- 155 (29.6%) non-respondent - **Measurement**: not reaching 18.5 kg/m^2	The factors that affect non-response were not assessed
Sadler, et al , [25] Ethiopia	Comparative follow up study. not peer reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each. 1956 and 639 were including in the final analysis	Health facilities / health centres **Food type**: RUFT/RUSF	- 1316 (67.4%) non-respondent in the intervention group and 519 (81.2) become non-respondent in the comparison group - **Measurement**: not reaching 18.5 kg/m^2	Factors that were associated with nutritional non-response were: - attending the program at a hospital was associated with an increased likelihood of non-response (OR=1.75, p=0.021), - severe food insecurity was associated with a decreased likelihood of non-response (OR=0.46, p=0.016) compared to those who were "food secure.
Tesfay et al , [42] Ethiopia	Retrospective cohort study	Adults living with HIV and-BMI<18.5	1757	Hospitals **Food type**: RUFT/RUSF	- 21% (95% CI = 19.7–23.4) completed the program but failed to recover from malnutrition. - **Measurement**: not reaching 18.5 kg/m^2	- urban residence, (AOR = 1.46, 95% CI = 1.4–2.91), - attending Shul (AOR = 2.92, 95% CI = 2.04–4.19) and - Lemlem Karl (AOR = 1.49, 95% CI 1.05–2.11) hospitals

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Default from the nutritional program

Two studies from Ethiopia reported on program default and factors associated with it.

In one study, 19% of the study participants did not complete the nutritional program. Factors that were associated with default were living in urban areas, being employed, attending at a hospital, and in clinical stages II and III [42]. In the other study, the default rate was 70.6% [25]. Being on ART for less than six months and being anaemic were associated with default [25] (**Table 6**).

In both studies, severe undernutrition at enrolment to the nutritional program was associated with default from the nutritional program [25 42].

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Table 6: Default from the nutritional program and its determinants in sub-Saharan Africa

Author, year, and country	Study design	Study population	N	Setting	Default to the nutritional program	Factors that contribute to default to the nutritional program
Sadler, [25] et al , Ethiopia	Comparative follows up study. not peer-reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 intervention were included in the final analysis	Health facilities / health centres	70.6% defaulted from the nutritional program among the intervention group	- severe acute malnutrition (OR: 1.89, P-value=0.003)
Tesfay et al [42] , Ethiopia	Retrospective cohort study	Adults living with HIV and BMI<18.5	1757	Hospitals	19% (95% CI, 17.3–20.7) did not complete the program	Factors associated with program incompleteness - living in urban areas (AOR = 1.44, 95% CI = 1.05– 1.97), - employed (AOR = 1.39, 95% CI = 1.01–1.93), - attending Shul (AOR = 4.6, 95% CI = 3.15–6.71) and Lemlem Karl (AOR = 2.5, 95% CI = 1.69–3.71) hospitals, - in clinical stages II (AOR = 2.49, 95% CI = 1.59–3.91) and III (AOR = 1.46(1.02–2.07), - on ART for less than six months (AOR = 1.61, 95% CI = 1.09–2.39), - anaemic (AOR = 1.77, 95% = 1.29–2.41), and - severe acute undernutrition at enrolment (AOR = 6.43, 95% CI = 4.69–8.3)

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Discussion

Given the susceptibility of people living with HIV to undernutrition, scaling up nutritional programs in HIV care in combination with other HIV services is vital [45]. Despite the provision of nutritional programs in HIV care settings in many countries [21], the existing evidence regarding the effectiveness of these programs is fragmented. This review synthesised the existing evidence regarding the impact of nutritional programs on the nutritional outcomes of people living with HIV.

Broadly, a range of anthropometric, clinical, and socioeconomic parameters was used as enrolment criteria in nutritional programs in the HIV care setting in sub-Saharan Africa. The duration of the nutritional programs was characterised by big variation from program to program. For instance, nutritional programs that aimed at the treatment of acute undernutrition were characterised by a shorter duration (ranging from three to six months) and provision of highly dense nutritional support (RUTF or RUSF) [25 38 42]. However, there were examples where nutritional support was provided for more than nine months [31 32 34]. The duration of nutritional programs that used corn-soy blended ranged from 14 weeks [32 34] to 9 months [37], and 12 weeks in programs that used the WFP food rations [35].

There was no optimal duration of the nutritional program because of the difference in content and nutritional value of the nutritional support used from program to program. Nutritional programs that use dense nutritional supports such as RUTF and RUSF may achieve faster weight gain and perhaps contribute to a shorter duration in the nutritional program [25 42]. These findings are consistent with outcomes from other studies conducted in other parts of the world such as India [46].

There is strong evidence in the literature regarding the role of nutritional programs in enhancing weight gain among those enrolled in a nutritional program with BMI<18.5 kg/m^2 [47]. However, in nutritional programs with predetermined program exit criteria, there was substantial variation in the magnitude of nutritional recovery. Nutritional recovery beyond attaining weight gain can vary from program to program depending on the availability of resources. In nutritional programs that used BMI>18.5kg/m^2 in Ethiopia, nutritional recovery was 32.6% [25], 55.3% [42] and 62.2% [44]. All other studies that did not use a defined cut-off point or different (e.g. one that used BMI>18 and other that used BMI>20) to determine nutritional outcome demonstrated that nutritional support of various forms are associated with better nutritional outcomes measured in other ways (improved BMI, weight gain, and MUAC). A systematic review that examined the impact of nutritional supplementation on HIV nutritional and clinical prognosis in sub-Saharan Africa found increased weight gain [48].

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Evidence from the current review showed that few studies reported on the rate of overall nutritional program failure. Two studies from Ethiopia found program failure rates involving non-response, default, and mortality rates that ranged from 37.6% [44] to 48.0%[38].

Studies examining non-response to the nutritional program (not reaching a BMI of 18.5kg/m^2 in two consecutive visits) were all from Ethiopia. While the nutritional program was similar in terms of the duration, type of food supplement, funding organisation, and policy, there was a big variation in the magnitude of non-response rate between these studies of 21% [42], 29.6% [44], and 67.4% [25]. The two studies [42] [44] with a relatively low non-response rate were conducted recently, after several years of implementation of the nutritional program in Ethiopia which might contribute to the low program non-response due to the accumulation of experience by the program.

Two studies from Ethiopia measured default from the nutritional program and the difference in default between the two studies was substantial −ranging from 19% [42] to 70.6 % [25]. While the study by Tesfay et al (2020) was conducted recently, the later study by Sadler (2012) was conducted at the initial stage of the nutritional program as a pilot evaluation which might also have contributed to a limited understanding of the program by implementers and beneficiaries. Furthermore, the study by Sadler (2012) may not have reflected the real program operation, which could have contributed to a higher magnitude of program non-response and default from the nutritional program [25]. Reasons for nutritional program ineffectiveness could include poor taste, dietary boredom, or HIV-related complications such as thrush as highlighted in a systematic review that involved resource adequate and constrained settings [49].

It is very difficult to compare nutritional outcomes by country because of the lack of uniformity in the nature of the nutritional program in terms of the duration, enrolment criteria, and the nutritional support that is given to people living with HIV. For instance, nutritional programs in Ethiopia used ready to use therapeutic or supplementary food for three months for mild and moderately malnourished and six months for severe malnutrition [25] [42] [44]. On the other hand, nutritional programs from Kenya used ready to use therapeutic or supplementary foods provided in a rural village but no specific duration was given for the nutritional program [40]. Nutritional recovery (achieving BMI greater than 20 kg/m^2) in Kenya [40] was lesser (13%) than studies from Ethiopia which were 62.2% [44], and 32.6% [25].

The studies also provided some insights into factors associated with program outcomes, including socio-demographic factors. Gender was highlighted, where being male was associated with nutritional program failure [38] and lower rates of recovery [40] [25] [44] and living in urban areas was associated with program non-response [42]

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In one study, attending the nutritional program in a hospital was associated with lower rates of nutritional recovery [25]. Another study found differences between hospitals where attending the nutritional program at relatively rural hospitals was associated with higher rates of non-response [42].

Clinical factors were also linked to program outcomes. For example, not starting ART was associated with nutritional program failure in one study [38], but another study found it was associated with nutritional program recovery [40]. This might be due to the fact that many individuals in the study by Ahoua et al 2011 were eligible for ART despite the lack of ART initiation because of delayed patient laboratory results or patient refusal. These patients might have died, giving rise to a high magnitude of program failure in the former study. Being on ART for less than six months were associated with default from the nutritional program [42]. The absence of opportunistic infection at enrolment was associated with nutritional recovery in one study [38].

Severe undernutrition at baseline was associated with a higher nutritional program failure rate [38], lower likelihood of achieving nutritional recovery [40], and default from the nutritional program [42]. On the other hand, mild and moderate undernutrition at enrolment was associated with nutritional recovery in one study [44]. Compared with those who are not enrolled in the nutritional intervention, those enrolled in the nutritional program were more likely to recover [25]. One study found that those with severe food insecurity were less likely to become non-respondents to the nutritional program [25].

Given the clinical, social, and economic susceptibility of people living with HIV to undernutrition [50], findings from the current review highlighted that nutritional programs in sub-Saharan Africa have focused on the treatment of undernutrition among people living with HIV [25] [38] [40]. For instance, terms such as nutritional recovery, weight gain, default, and nutritional recovery were used in all of the studies included to imply the short term positive impact of the nutritional programs among people living with HIV. Measures used in all nutritional programs were clinical and none provided measures on longer-term nutritional wellbeing. Longer-term measures such as food security were not included as key program goals in any of the nutritional programs despite the wider vulnerability of people living with HIV to food insecurity in sub-Saharan Africa [51] [52].

Strength and Limitations

The key strength of this study is because it has included studies from a wider geographic location. Some studies had a very small sample size (e.g. [41] (n=35) and [43] (n=106)) which may negatively impact the estimates of the nutritional outcomes and factors associated with

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them. It was not possible to pool estimates of nutritional outcomes because of the differences in enrolment and exit criteria and program duration between nutritional programs.

Conclusion and recommendations

Despite the use of various enrolment and exit criteria, nutritional programs in HIV care have led to overall improvements in the nutritional outcomes of people living with HIV. However, the programs reviewed were characterised by problems of effectiveness such as a high magnitude of default and non-response. Various sociodemographic, clinical, and nutritional characteristics have an impact on the nutritional outcomes, which gives potential avenues for future research and program improvement. However, the goal of most nutritional programs in HIV care settings is to reverse short term nutritional problems such as undernutrition. In contrast, nutritional program effectiveness in HIV care should be measured in terms of the program's capacity to sustainably address the nutritional wellbeing of individual HIV patients and others around them over the longer term [53].

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Author contributions: **FT** conceptualised, designed the study, conducted the search, analysed, and interpreted the data as well-drafted the manuscript. **HAG** assisted with the analysis. **SJ, AZ, HAG, and LM** critically reviewed the design, analysis, and interpretation of the findings and critically reviewed and approved the manuscript.

Any checklist and flow diagram: The PRISMA diagram was used.

A patient consent form: All primary studies were assessed for the fulfillment of ethical standards and all fulfilled the ethical standards.

Data statement: The authors are happy to share the data extracted from the included papers.

Supplementary and raw data: We do not have supplementary raw data but we are happy to share the data extracted from each included paper.

Acknowledgments: This manuscript builds on a Ph.D. thesis, and we would like to acknowledge the Australian Government Research Training program and Flinders University for supporting the study through a student support payment and fieldwork funding. We would also like to acknowledge Flinders University librarian−Shannon Brown for assisting the searching process of this systematic review.

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Supplementary file 2: Assessment of methodological quality (n=16).

Authors	Q	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	%
Ahoua, et al. (2011)	Y	N/A	N/A	Y	Y	Y	Y	Y	Y	100
Bahwere, et al. (2009).	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Berhe, et al. (2014)	Y	N/A	N/A	N	Y	Y	Y	Y	Y	86
Diouf, et al. (2016).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Mallewa, et al. (2018).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Manary, et al. (2010)	Y	Y	Y	Y	Y	Y	Y	Y	U	100
Nagata, et al. (2014)	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Ndekha, et al. (2009)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Oguntibeju,et al. (2007)	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Olsen, et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Rawat, R., et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Rawat, et al. (2010).	Y	N/A	N/A	N	Y	Y	Y	Y	Y	86
Sadler, et al. (2012).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Scarcella, et al. (2011)	Y	Y	Y	N	Y	Y	Y	Y	Y	89
Tesfay, et al. (2020).	Y	Y	Y	N	Y	Y	Y	Y	Y	89
van Oosterhout, et al.	Y	Y	Y	Y	Y	Y	Y	Y	Y	100

Q= Question Y=Yes; N= No; NA= Not applicable, UC=unclear

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Supplementary file 3: JBI Critical Appraisal Checklist for Quasi-Experimental Studies (non-randomized experimental studies)

Reviewer_____Date_____

Author_____Year_____Record Number _____

	Yes	No	Unclear	Not applicable
1. Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	☐	☐	☐	☐
2. Were the participants included in any comparisons similar?	☐	☐	☐	☐
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	☐	☐	☐	☐
4. Was there a control group?	☐	☐	☐	☐
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	☐	☐	☐	☐
6. Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	☐	☐	☐	☐
7. Were the outcomes of participants included in any comparisons measured in the same way?	☐	☐	☐	☐
8. Were outcomes measured in a reliable way?	☐	☐	☐	☐
9. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

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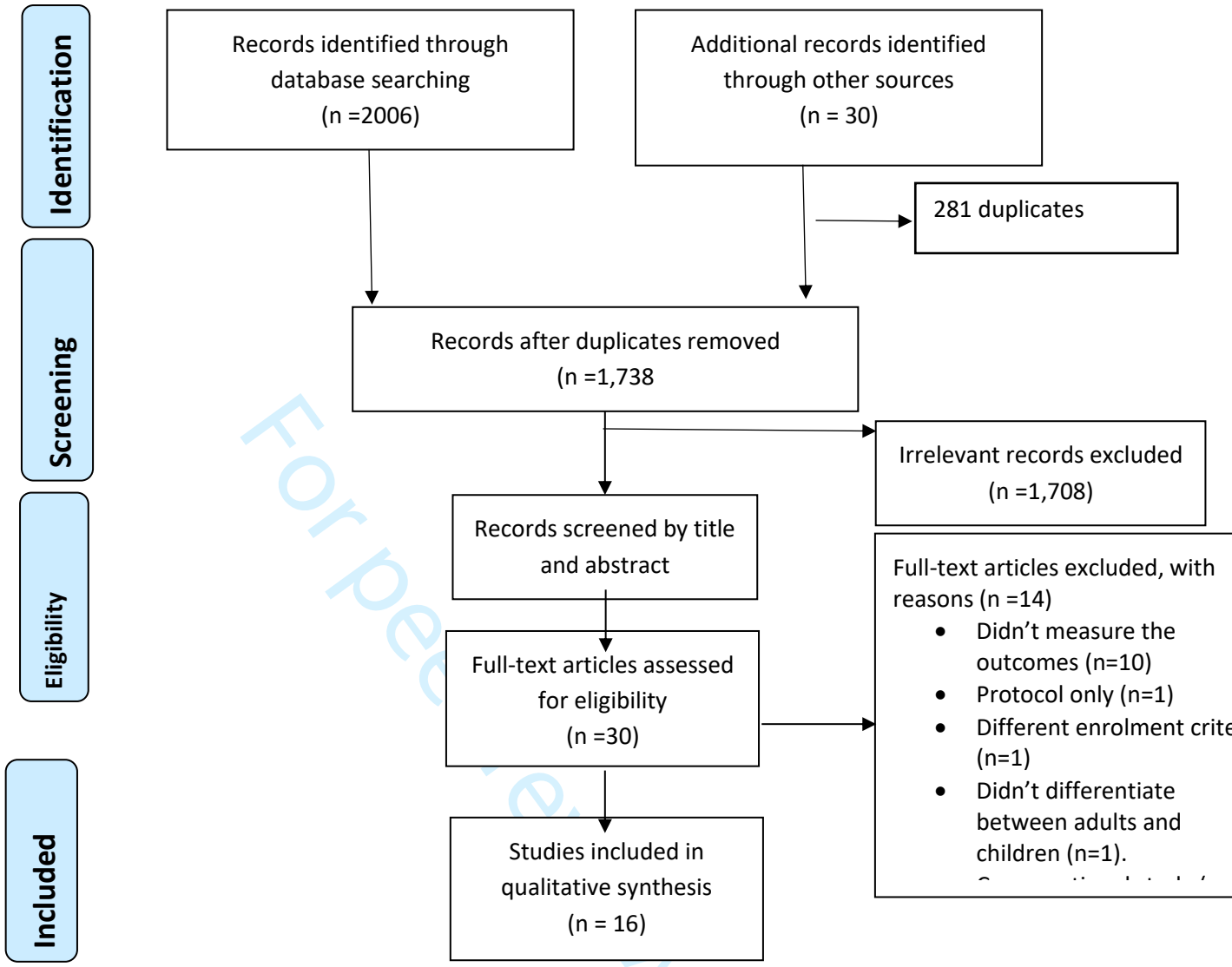


Figure 1: PRISMA flow chart

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Supplementary file 1: Shows the searching terms and outputs for Medline

#	Searches	Results
1	HIV Infections/ or HIV/ or HIV-2/ or HIV-1/	229102
2	Acquired Immunodeficiency Syndrome/	75776
3	(HIV* or AIDS* or "Acquired Immunodeficiency Syndrome").tw,kf.	382368
4	1 or 2 or 3	420727
5	Nutritional Status/	41768
6	Dietary Supplements/	52953
7	Food Assistance/ or *Food/ or *"Diet, Food, and Nutrition"/ or Food, Fortified/	28061
8	nutrition therapy/ or diet therapy/ or nutritional support/	18224
9	((food or nutrition) adj3 (support or program or intervention or supplement* or therap* or care or consult*)).tw,kf.	24448
10	5 or 6 or 7 or 8 or 9	151790
11	"africa south of the sahara"/ or africa, central/ or cameroon/ or central african republic/ or chad/ or congo/ or "democratic republic of the congo"/ or equatorial guinea/ or gabon/ or "sao tome and principe"/ or africa, eastern/ or burundi/ or djibouti/ or eritrea/ or ethiopia/ or kenya/ or rwanda/ or somalia/ or south sudan/ or sudan/ or tanzania/ or uganda/ or africa, southern/ or angola/ or botswana/ or lesotho/ or malawi/ or mozambique/ or namibia/ or south africa/ or swaziland/ or zambia/ or zimbabwe/ or africa, western/ or benin/ or burkina faso/ or cabo verde/ or cote d'ivoire/ or gambia/ or ghana/ or guinea/ or guinea-bissau/ or liberia/ or mali/ or mauritania/ or niger/ or nigeria/ or senegal/ or sierra leone/ or togo/	197050
12	("africa south of the sahara" or africa*or cameroon or central african republic or chad or congo or "democratic republic of the congo" or equatorial guinea or gabon or "sao tome and principe" or burundi or djibouti or eritrea or ethiopia or kenya or rwanda or somalia or south sudan or sudan or tanzania or uganda or angola or botswana or lesotho or malawi or mozambique or namibia or "south aftrica" or swaziland or zambia or zimbabwe or benin or burkina faso or "cabo verde or cote d'ivoire" or gambia or ghana or guinea or guinea-bissau or liberia or mali or mauritania or niger or nigeria or senegal or sierra leone or togo).mp.	361692
13	11 or 12	412824
14	4 and 10 and 13	616
15	limit 14 to english language	604
16	limit 15 to yr="2005 -Current"	510

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Supplementary file 2: JBI Critical Appraisal Checklist for Quasi-Experimental Studies (non-randomized experimental studies)

Reviewer_____Date_____

Author_____Year_____Record Number _____

	Yes	No	Unclear	Not applicable
1. Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	☐	☐	☐	☐
2. Were the participants included in any comparisons similar?	☐	☐	☐	☐
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	☐	☐	☐	☐
4. Was there a control group?	☐	☐	☐	☐
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	☐	☐	☐	☐
6. Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	☐	☐	☐	☐
7. Were the outcomes of participants included in any comparisons measured in the same way?	☐	☐	☐	☐
8. Were outcomes measured in a reliable way?	☐	☐	☐	☐
9. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

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Supplementary file 3: Assessment of methodological quality (n=16).

Authors	Q	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	%
Ahoua, et al. (2011)	Y	N/	N/A	Y	Y	Y	Y	Y	Y	10
Bahwere, et al. (2009).	Y	N/	N/	UC	Y	Y	Y	Y	Y	10
Berhe, et al. (2014)	Y	N/	N/A	N	Y	Y	Y	Y	Y	86
Diouf, et al. (2016).	Y	Y	Y	Y	Y	Y	Y	Y	Y	10
Mallewa, et al. (2018).	Y	Y	Y	Y	Y	Y	Y	Y	Y	10
Manary, et al. (2010)	Y	Y	Y	Y	Y	Y	Y	Y	U	10
Nagata, et al. (2014)	Y	N/	N/A	UC	Y	Y	Y	Y	Y	10
Ndekha, et al. (2009)	Y	Y	Y	Y	Y	Y	Y	Y	Y	10
Oguntibeju,et al. (2007)	Y	N/	N/A	UC	Y	Y	Y	Y	Y	10
Olsen, et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	10
Rawat, R., et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	10
Rawat, et al. (2010).	Y	N/		N	Y	Y	Y	Y	Y	86
Sadler, et al. (2012).	Y	Y	Y	Y	Y	Y	Y	Y	Y	10
Scarcella, et al. (2011)	Y	Y	Y	N	Y	Y	Y	Y	Y	89
Tesfay, et al. (2020).	Y	Y	Y	N	Y	Y	Y	Y	Y	89
van Oosterhout, et al.	Y	Y	Y	Y	Y	Y	Y	Y	Y	10

Q= Question Y=Yes; N= No; NA= Not applicable, UC=unclear

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	N/A

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	N/A
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	26
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	30
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	30
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	No funding

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

Characteristics and impacts of nutritional programs to address undernutrition of adults living with HIV in sub-Saharan Africa: A systematic review of evidence.

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-047205.R1
Article Type:	Original research
Date Submitted by the Author:	25-Mar-2021
Complete List of Authors:	Tesfay, Fisaha; Mekelle University College of Health Sciences, School of Public Health Javanparast, Sara; Flinders University, College of Medicine and Public Health Gesesew, Hailay ; Flinders University, College of Medicine and Public Health Mwanri, Lillian; Flinders University, College of Medicine and Public Health Ziersch, Anna; Flinders University, Southgate Institute for Health, Society and Equity
Primary Subject Heading:	HIV/AIDS
Secondary Subject Heading:	Epidemiology, Health services research, HIV/AIDS
Keywords:	PUBLIC HEALTH, NUTRITION & DIETETICS, EPIDEMIOLOGY, Infectious disease/HIV < NEUROLOGY

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Characteristics and impacts of nutritional programs to address undernutrition of adults living with HIV in sub-Saharan Africa: A systematic review of evidence.

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Abstract

Objectives: Although some studies have identified various challenges affecting nutritional programs to effectively tackle undernutrition amongst people living with HIV, evidence about the characteristics and impacts of these programs on nutritional outcomes varies based on the country's contexts, specific program goals and the implementation processes. This systematic review sought to synthesise evidence on the characteristics and impact of nutritional programs on nutritional outcomes of people living with HIV in sub-Saharan Africa.

Design: Systematic review

Data sources: We searched for primary studies published in the following databases: Web of Science, Medline, Scopus, ScienceDirect, ProQuest and Google Scholar, supplemented by checking reference lists of included papers.

Eligibility criteria: Studies published from 2005 to the 10th of July 2020 and reporting on the nutritional outcomes of undernourished people enrolled in nutritional programs in HIV care in Sub-Saharan Africa.

Data extraction and synthesis: Data were extracted using a data extraction proforma. Nutritional outcomes of people living with HIV before and after enrolment in a nutritional program were compared and narratively synthesised.

Results: Sixteen studies assessing the impact of nutritional programs in HIV care on nutritional outcomes were included**.** Of these, 13 examined nutritional programs implemented in health facilities and the remaining three were delivered outside of health facilities. Nutritional recovery (defined differently in the studies) ranged from 13.1% to 67.9%. Overall program failure rate, which included default after enrolment in a nutritional program or non-response, ranged from 37.6% to 48.0%. More specifically, non-response to a nutritional program ranged from 21.0% to 67.4% and default from the program ranged from 19.0% to 70.6%. Key sociodemographic, clinical, and nutritional characteristics that affect nutritional recovery, non-response and default were also identified.

Conclusions and recommendations: Nutritional programs in HIV care have led to some improvements in improved nutritional outcomes amongst people living with HIV. However, the programs were characterised by a high magnitude of default and non-response. To improve desired nutritional outcomes of people living with HIV, a holistic approach that addresses longer-term determinants of undernutrition is needed.

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Keywords: Nutritional program, nutritional support, HIV and nutrition

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Strengths and limitations

- This review is the first of its kind to synthesise the evidence with regards to outcomes of nutritional programs to address undernutrition for people with HIV across sub-Saharan Africa, which has the largest number of people living with HIV.
- The review assesses multiple nutritional outcomes and presents a summary of evidence from a range of nutritional programs.
- However, the diversity and heterogeneity of the studies and nutritional programs meant that meta-analysis was not possible to be done, a shortcoming of the review.
- This study only focused on papers published in English which may have excluded potential studies on the same topic but published in other languages.

The original protocol for the study: The study protocol is attached at the end of the study.

Funding statement: Not funded.

A competing interest statement. All authors declare that they have no conflict of interest.

Introduction

Evidence demonstrates the clinical and social susceptibility of people living with HIV to undernutrition and its wider effects [1] [2]. Inadequate food intake due to lack of appetite, malabsorption, and impaired storage and metabolism are clinical causes of undernutrition among people living with HIV [3], while social factors such as poverty, gender inequality, food insecurity, stigma, and discrimination also hinder access to adequate food and nutrition [4] [5]. Undernutrition has various impacts on the wellbeing of people living with HIV, including on HIV-related mortality in resource poor settings which has been highlighted in a range of studies [6-8]. Research has also indicated that undernutrition can act as a barrier to the utilisation of HIV care services and antiretroviral therapy (ART) adherence [9-11].

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The World Health Organisation (WHO) highly recommends the integration of nutritional interventions into HIV care and support services [12 13]. To address the impact of undernutrition among people living with HIV, nutritional programs targeting people living with HIV are operational in resource-poor settings, mainly in sub-Saharan Africa [14]. The key elements of these programs are nutritional assessment, counselling, and nutritional support [15-17]. The latter component includes the provision of micronutrient, macronutrient, or fortified macronutrient supplementation [18 19].

Nearly 90% of HIV care sites in sub-Saharan Africa have responded to the WHO recommendations and instituted some forms of nutritional program [20], with substantial variations between and within countries [16 18 21]. Donor organisation policies regarding the nutritional program and availability of national and local resources have led to program differences observed between countries in sub-Saharan Africa [22], including the type of care [23-25] and nature of nutritional support provided [26-28]. Evidence regarding the impact of nutritional programs on nutritional outcomes is fragmented, mixed, and varies by country context, program goals, and implementation. This study aims to systematically review and synthesise evidence on the characteristics and impact of nutritional programs on nutritional outcomes of undernourished people enrolled in nutritional programs in HIV care in sub-Saharan Africa.

Methods

Protocol

The protocol for this systematic review has been registered under the PROSPERO review registration number: CRD42020196827.

Systematic search strategy

This review was guided by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). Studies published in English language in the Web of Science, Medline, Scopus, ScienceDirect, ProQuest, from 2005 to the 10th of July 2020 were searched. Google Scholar and the reference lists of references of selected papers were also searched. The search concepts and terms included “Nutritional programs”, “HIV/AIDS”, and “sub-Saharan Africa” (**Table 1**). Outputs of Ovid Medline are shown in the **supplementary file 1**. **Table 1:** Concepts and their respective keywords

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Concept 1: ***Nutritional program***	**Concept 2:** ***HIV/AIDS***	**Concept 3:** ***Sub-Saharan Africa***
Nutritional program	HIV-1	South of Saharan Africa
Nutritional intervention	HIV-2	Africa
Nutritional care	HIV infection	Western Africa
Food by prescription	Human immunodeficiency syndrome	Central Africa
Supplementary food	Acquired immune deficiency syndrome	
Ready to use Therapeutic food		
Food supplementation	AIDS	List of all countries in sub-Saharan Africa

Selection of studies

The screening process involved two stages. The first stage was conducted by FT and involved screening the studies by title and abstracts against the inclusion criteria. The second stage was conducted by two independent reviewers (FT and HAG) and involved full-text screening of articles. The inclusion criteria were:

- longitudinal studies (cohort and interventional) that examined nutritional outcomes of undernourished people (age>15 years) who were enrolled in nutritional programs in HIV care settings.
- studies measuring nutritional outcomes using Body Mass Index, (BMI), weight gain or Mid Upper Arm Circumference (MUAC)
- studies published since 2005, because most nutritional programs in sub-Saharan Africa started in 2005 and published in English.

Studies involving children and pregnant women were excluded because different enrolment and nutritional outcome criteria are used to enrol children and pregnant women in programs. Studies were also excluded from the review if information about age or pregnancy status was not provided (**Figure 1**).

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Measurement of exposures

The exposure variables included:

- sociodemographic factors: age, sex, residence, marital status, and educational status.
- clinical factors: WHO clinical stage, CD4 count, functional status, presence of opportunistic infections, and HIV status disclosure.
- nutritional factors: baseline nutritional status, the status of food security and other dietary information.

Measurement of outcomes

The outcome variables were nutritional outcomes of undernourished adults living with HIV, compared before and after enrolment in a nutritional program. Outcomes included changes in BMI, weight, or MUAC and nutritional recovery (BMI$\geq$ 18.5 kg/m^2), default, or non-response to the nutritional program.

Context/setting

Most nutritional programs in sub-Saharan Africa commenced in 2005. Most nutritional programs in sub-Saharan Africa included in the reviewed studies had similar program components which included nutritional assessment, counselling, and nutritional support. This review is limited to the sub-Saharan Africa given the duration, enrolment, and exit criteria of nutritional programs are very different in other regions and settings.

Assessment of methodological quality

The Joanna Briggs Institute [29] for critical appraisal tool for quasi-experimental studies was used to assess the quality of studies that were included in the review. Two independent reviewers (FT and HAG) conducted the quality assessment.

Data extraction

Data were extracted using a proforma adopted from JBI-MAStARI. The extracted data involved information about the outcome and exposure that affected the nutritional outcomes. In addition, population, sample size, country description of the nutritional program, and the conclusion of the studies were extracted.

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Data analysis

Due to the heterogeneity of the studies included in terms of study design, the measure of nutritional outcomes used, and differences in enrolment and exit criteria, a narrative synthesis was the most appropriate analysis procedure for this review. Different variables were identified from the reported findings of each study. Similar patterns were grouped in subsequent coding and analysis performed. Finally, the findings were categorised into two major themes.

Patient and Public Involvement: No patient involved.

Ethics approval statement: Ethics approval was not relevant in this study because it is a systematic review, but we have ensured all studies included in this study has ethically approved.

Results

Description of the studies

A total of 2036 papers were obtained from electronic databases and other sources. After removing duplicates 1,738 studies were screened by title and abstract and 1,708 were deemed irrelevant. The remaining 30 papers were scrutinised further against the inclusion criteria using full-text screening. Sixteen papers involving 25,335 participants were included in the final review (**Image 1**). Of the 16 final studies included in the review, nine were retrospective cohort studies, three were prospective cohort studies, and four were randomised trials.

Assessment of methodological quality

The methodological and reporting quality of the reviewed studies was assessed using the JBI critical appraisal tool for quasi-experimental studies (**supplementary file 2**). The assessment tool included nine parameters each carrying equivalent weight (**supplementary file 3**).

Nine studies, six clinical trials [30-35], and three cohort studies [25] [36] [37] scored nine out of nine (100%).

One retrospective cohort study [38] scored seven out of nine but two parameters (related to the comparability of study participants and similarity of interventions in both the exposed and non-exposed groups) were not applicable− scoring 100%. Three cohort studies scored six out of nine which is 100% as the same two parameters were not applicable, and the presence of control groups was not clear [39-41].

Two retrospective studies scored eight out of nine (89%) as there was no control group [42] [43]. Another retrospective cohort study scored six out of nine, with no control group and two parameters not applicable - a final score of 86% [44].

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Characteristics of the nutritional programs

The nutritional programs examined in this review varied in terms of enrolment criteria, duration, and type of nutritional support used. Six studies used BMI <18.5 kg/m^2 as an enrolment criteria [25 33 34 40 42 44]. Another seven studies used one of the following: bilateral oedema, being bedridden for 50% of the month, TB coinfection, CD4 count < 350, WHO clinical stage III and IV, in addition to BMI< 18.5 kg/m^2 or MUAC <210mm) [31 32 37-39 41 43]. Living in a household with HIV positive family members was used as an enrolment criterion in one study [36]. Another study enrolled people living with HIV at any stage of the disease and not able to eat [30]. A further study used the world food program (WFP) poverty assessment index, not receipt of other food assistance, CD4 count between 200 and 450 cells per microliter, and ART naïve (new for the treatment) as an enrolment criterion [35].

In five studies, the duration of the nutritional program ranged from three months (for mild and moderate malnutrition) to six months (for severe malnutrition) [25 38 39 42 44]. Two studies reported nine weeks of program duration [33 44], two had 14 weeks [32 34], and a further study [31] 24 weeks duration before individual patients exited from the program. The nutritional program was nine months in one study [37] and twelve months in three studies [33 35 36]. No duration limit was reported in two studies [40 43].

Various types of nutritional supports were used in nutritional programs. Ready to use therapeutic food (RUTF) or ready to use supplementary food (RUSF) were used in eight studies [25 30 31 33 38 39 42-44]. Corn/soy blended food was used in three studies [32 34 37 40] and the WFP food basket (which involves generic food rations) was used in two studies [35 36]. One study used an extract of hypoxis, grapefruit seed extract, sitosterol and sitosterolin, beta-carotene, vitamin E, vitamin, vitamin B1, vitamin B2, vitamin B12, nicotinamide, vitamin C, olive green leaf extract, folic acid, and natural antioxidant (biocydin) as the nutritional supplement [41].

Although nutritional assessment was an element of nutritional programs in all studies included in this review, the provision of nutritional counselling was stated only in two studies [33, 42]

The nutritional programs were provided in health facilities such as hospitals and clinics in 12 studies [25 30-34 36-38 42-44], in community-based settings in four studies [33 39 40], and no setting was specified in one study [41].

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Impact of the nutritional program on nutritional outcomes

In reporting improvements in the nutritional outcome of people living with HIV, studies used various outcome measures. Gain in BMI, weight, and MUAC were used either in combination or as separate nutritional outcome measures. BMI was used in nine studies [25 34 35 37 38 41-44] and both MUAC and BMI were used in two studies [31 39]. The other two studies used weight gain [30 36], and three studies used both BMI and weight gain to measure the nutritional outcomes [30 33 36].

Nutritional program failure, recovery, default, and non-response were main outcomes measured in most review studies. The definition of these terms in the studies included are shown in **table 2.**

Table 2: The definition of nutritional outcomes in nutritional programs

Outcomes	**Adults**
Program failure	Includes default and non-response from the nutritional program
Improvement in nutritional status and recovery /graduation from the nutritional program	Various criteria were used to determine nutritional recovery • Reaching BMI of 18.5kg/m^2 for two consecutive visits in some studies or BMI>18 kg/m^2 or BMI>20kg/m^2 In other studies, the nutritional outcome was measured in terms of improving weight, BMI, or MUAC.
Non-response	Participants did not reach the target BMI for two consecutive visits or did not show any change in BMI, weight, and MUAC
Default /failure to complete the nutritional program	Dropped out of the program before reaching the target BMI, weight or MUAC

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Improvement in nutritional status or recovery from the nutritional program

Eleven studies measured nutritional recovery which was defined differently in various nutritional programs. Three studies defined program recovery as BMI>18.5kg/m^2 in two consecutive measurements [25 42 44], and nutritional recovery in these studies ranged from 32.6% [25] to 62.2% [44]. One study used BMI >18 kg/m^2 in two subsequent measurements and the absence of oedema, and recovery was 47.4% [38]. A further study defined program recovery as BMI>20kg/m^2 and 13.1% attained nutritional recovery against these criteria [40].

Four studies used improvements in BMI, without specifying a specific cut-off point [34 35 37 43]. The improvement in BMI in these studies ranged from 0.6 kg/m^2 using WFP rations for 12 months [35] to 2.2 kg/m^2 in a study where participants received a fortified spread for 14 weeks [34] and those who used a corn-soy blend showed a BMI increase of 1.7 kg/m^2 [34]. Weight gain was used to measure nutritional outcomes in four studies [30 33 36 39] and weight gain ranged from 0.85 kg [33] to 3 kg [39].

One study using weight gain found that after 9 weeks of supplementation, body weight, and fat-free mass increased significantly by 11 % (p= 0.033), and 11.8 % (p= 0.033) in the RUTF group, but not in the control group [30]. One study used BMI, weight gain, and MUAC [31], and those in the RUSF group had significantly greater gains in weight, BMI, and MUAC than those in the no-RUSF group (p=0·004, and 0·03, respectively).

Annex table 1 presents the socioeconomic, demographic, clinical, and nutritional determinants of nutritional outcomes among those enrolled in nutritional program. For sociodemographic factors that affected nutritional outcomes, female gender was associated with nutritional recovery in three studies [44 25 40]. Compared with those who didn't attend any schooling, individuals who attended school were also more likely to recover than those who did not in one study [44], and attending the nutritional program in a hospital was inversely related to nutritional recovery in another study [25].

Few clinical characteristics were associated with nutritional recovery. In one study, not being on ART was associated with nutritional recovery [40]. In another study, compared with WHO clinical stage 4, being at WHO clinical stage 1 and 2 were associated with nutritional recovery [44]. In addition, the absence of opportunistic infection at enrolment to the nutritional program was associated with nutritional recovery compared with those who had an opportunistic infection [44].

Regarding the impact of nutritional characteristics at enrolment, compared with those enrolled with severe undernutrition, those with moderate undernutrition were associated with nutritional

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recovery in two studies [40] [44]. In one study, enrolment in the nutritional program with mild undernutrition was associated with nutritional recovery compared with those with severe undernutrition [44].

Program failure rate

The program failure rate involves a combination of default, non-response, and mortality while enrolled in the nutritional program. The program failure rate was reported in two studies [38] [44]. The nutritional program failure rate in a study from Kenya and Uganda was 48% [38] and 37.6% in a study from Ethiopia [44]. Factors that were associated with the program failure were only assessed in one study [38] which found male gender, severe undernutrition at program enrolment (pre-ART), and ineligibility for ART at enrolment as variables associated with program failure (**Annex table 2**).

Non-response from the nutritional program

Three studies from Ethiopia measured non-response to the nutritional program ranging from 21% [42], to 29.6% [44], and 67.4% [25].

Compared with those who attended primary health care services to receive nutritional services, attending a hospital was associated with greater non-response in one study [25]. A study by Tesfay and colleagues examining health facility based nutritional programs in Ethiopia reported that being an urban area was associated with non-response [42].

In terms of clinical characteristics, the same study found advanced WHO clinical stage (stage IV), and bedridden functional status were associated with non-response to the nutritional program [42]. Compared with those enrolled with mild undernutrition, severe undernutrition at enrolment was also associated with non-response [42].

In another study, those with severe food insecurity were less likely to not respond than those who were food secure [25]. The third study did not assess factors contributing to non-response (**Annex table 3**).

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Default from the nutritional program

Two studies from Ethiopia reported program default and associated factors.

In one study, 19% of the study participants did not complete the nutritional program. Factors that were associated with default were living in urban areas, being employed, attending at a hospital, and in clinical stages II and III [42]. In another study, the default rate was 70.6% [25]. Being on ART for less than six months and being anaemic were associated with default [25] (**Annex table 4**).

In both studies, severe undernutrition at enrolment to the nutritional program was associated with default from the nutritional program [25] [42].

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Discussion

Given the susceptibility of people living with HIV to undernutrition, and the synergistic effect of undernutrition in accelerating HIV infection to AIDS, scaling up nutritional programs in HIV care in combination with other HIV services is vital [45]. Despite the provision of nutritional programs in HIV care settings in many countries [21], evidence regarding the effectiveness of these programs is fragmented–this review sought to address this with regards to sub-Saharan Africa.

Broadly, a range of anthropometric, clinical, and socioeconomic parameters were used as enrolment criteria in nutritional programs in the HIV care setting in sub-Saharan Africa. The duration of the nutritional programs was characterised by a significant variation between programs. For instance, nutritional programs that aimed at the treatment of acute undernutrition were characterised by a shorter duration (ranging from three to six months) and provision of highly dense nutritional support (RUTF or RUSF) [25 38 42]. However, there were examples where nutritional support was provided for more than nine months [31 32 34]. The duration of nutritional programs that used corn-soy blended ranged from 14 weeks [32 34] to 9 months [37], and 12 weeks in programs that used the WFP food rations [35].

There was no optimal duration of the nutritional program because of the difference in content and nutritional value of the nutritional support used from program to program. Nutritional programs that use dense nutritional supports such as RUTF and RUSF may achieve faster weight gain and perhaps contribute to a shorter duration in the nutritional program [25 42]. These findings are consistent with outcomes from other studies conducted in other parts of the world such as India [46].

There is strong evidence in the literature regarding the role of nutritional programs in enhancing weight gain among those enrolled in a nutritional program with BMI<18.5 kg/m^2 [47]. However, in nutritional programs with predetermined program exit criteria, there was substantial variation in the magnitude of nutritional recovery. Nutritional recovery beyond attaining weight gain can vary from program to program depending on the availability of resources. In nutritional programs that used BMI>18.5kg/m^2 in Ethiopia, nutritional recovery was 32.6% [25], 55.3% [42] and 62.2% [44]. Other studies did not use a defined cut-off point or used different BMI measure (e.g., one used BMI>18 and another used BMI>20) or other measures (weight gain, MUAC) to determine nutritional outcomes. All of these demonstrated that nutritional support of various forms were associated with better nutritional outcomes.

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A systematic review that examined the impact of nutritional supplementation on HIV nutritional and clinical prognosis in sub-Saharan Africa found increased weight gain [48].

Evidence from the current review showed that few studies reported on the rate of overall nutritional program failure. Two studies from Ethiopia found program failure rates involving non-response, default, and mortality rates that ranged from 37.6% [44] to 48.0%[38].

Studies examining non-response to the nutritional program (not reaching a BMI of 18.5kg/m^2 in two consecutive visits) were all from Ethiopia. While the nutritional program was similar in terms of the duration, type of food supplement, funding organisation, and policy, there was a big variation in the magnitude of non-response rate between these studies of 21% [42], 29.6% [44], and 67.4% [25]. The two studies [42] [44] with a relatively low non-response rate were conducted recently, after several years of implementation of the nutritional program in Ethiopia, which may suggest that program non-response rates may have dropped due to the accumulation of experience by the program.

Two studies from Ethiopia measured default from the nutritional program and the difference in default between the two studies was substantial −ranging from 19% [42] to 70.6 % [25]. While the study by Tesfay et al (2020) was conducted recently, the later study by Sadler (2012) was conducted at the initial stage of the nutritional program as a pilot evaluation and may reflect a less developed version of the program. Reasons for nutritional program ineffectiveness could include poor taste of the supplement, dietary boredom, or HIV-related complications such as thrush as highlighted in a systematic review that involved resource adequate and constrained settings [49].

It is very difficult to compare nutritional outcomes by country because of the lack of uniformity in the nature of the nutritional program in terms of the duration, enrolment criteria, and the nutritional support that is given. For instance, nutritional programs in Ethiopia used ready to use therapeutic or supplementary food for three months for mild and moderately malnourished and six months for severe malnutrition [25] [42] [44]. On the other hand, nutritional programs from Kenya used ready to use therapeutic or supplementary foods provided in a rural village but no specific duration was given for the nutritional program [40]. Nutritional recovery (achieving BMI greater than 20 kg/m^2) in Kenya [40] was lesser (13%) than studies from Ethiopia which were 62.2% [44], and 32.6% [25].

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The studies provided some insights into factors associated with program outcomes, including socio-demographic factors. Gender was highlighted, where being male was associated with nutritional program failure [38] and lower rates of recovery [40] [25] [44] and living in urban areas was associated with program non-response [42]. In one study, attending the nutritional program in a hospital was associated with lower rates of nutritional recovery [25]. Another study found differences between hospitals where attending the nutritional program at relatively rural hospitals was associated with higher rates of non-response [42].

Clinical factors were also linked to program outcomes. For example, not starting ART was associated with nutritional program failure in one study [38], but another study found it was associated with nutritional program recovery [40]. This might be due to the fact that many individuals in the study by Ahoua et al 2011 were eligible for ART despite the lack of ART initiation because of delayed patient laboratory results or patient refusal. These patients might have died, giving rise to a high magnitude of program failure in the former study. Being on ART for less than six months were associated with default from the nutritional program [42]. The absence of opportunistic infection at enrolment was associated with nutritional recovery in one study [38].

Severe undernutrition at baseline was associated with a higher nutritional program failure rate [38], lower likelihood of achieving nutritional recovery [40], and default from the nutritional program [42]. On the other hand, mild and moderate undernutrition at enrolment was associated with nutritional recovery in one study [44]. Compared with those who are not enrolled in the nutritional intervention, those enrolled in the nutritional program were more likely to recover [25]. One study found that those with severe food insecurity were less likely to become non-respondents to the nutritional program [25].

Given the clinical, social, and economic susceptibility of people living with HIV to undernutrition [50], findings from the current review highlighted that nutritional programs in sub-Saharan Africa have focused on the treatment of undernutrition among people living with HIV [25] [38] [40]. For instance, terms such as nutritional recovery, weight gain, default, and nutritional recovery were used in all of the studies included to imply the short term positive impact of the nutritional programs among people living with HIV. Measures used in all nutritional programs were clinical and none provided measures on longer-term nutritional wellbeing. Longer-term measures such as food security were not included as key program goals in any of the nutritional programs despite the wider vulnerability of people living with HIV to food insecurity in sub-Saharan Africa [51] [52].

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Strength and Limitations

The key strength of this study is its coverage of sub-Saharan Africa, a major area of HIV infection, and its review of multiple nutrition programs representing a large population base. It also examines a range of nutrition outcomes. However, the review has some limitations., Some studies had a very small sample size (e.g. [41] (n=35) and [43] (n=106)) which may negatively impact the estimates of the nutritional outcomes and factors associated with them. It was not possible to pool estimates of nutritional outcomes because of the differences in enrolment and exit criteria and program duration between nutritional programs. This study only included papers published in English which automatically excluded papers published in other languages, which may lead to reporting bias. The focus of this review is on the impact of nutritional programs on undernourished people enrolled in nutritional programs in HIV care settings. However, overnutrition has also been highlighted as an important issue for people living with HIV [16] and an examination of programs designed to address this would be an important area for future reviews.

Conclusion and recommendations

Despite the use of various enrolment and exit criteria, nutritional programs in HIV care have led to overall improvements in the nutritional outcomes and overall wellbeing of people living with HIV. However, the programs reviewed were characterised by problems of effectiveness such as a high magnitude of default and non-response. Various sociodemographic, clinical, and nutritional characteristics have an impact on the nutritional outcomes, which gives potential avenues for future research and program improvement. However, the goal of most nutritional programs in HIV care settings is to reverse short-term nutritional problems such as undernutrition. In contrast, the impact of nutritional programs in HIV care should be measured in terms of the program's capacity to sustainably address the nutritional wellbeing of individual HIV patients and others around them over the longer term [53].

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Author contributions: **FT** conceptualised, designed the study, conducted the search, analysed, and interpreted the data as well-drafted the manuscript. **HAG** assisted with the analysis. **SJ, AZ, HAG, and LM** critically reviewed the design, analysis, and interpretation of the findings and critically reviewed and approved the manuscript.

Any checklist and flow diagram: The PRISMA diagram was used.

A patient consent form: All primary studies were assessed for the fulfillment of ethical standards and all fulfilled the ethical standards.

Data statement: The authors are happy to share the data extracted from the included papers.

Supplementary and raw data: We do not have supplementary raw data but we are happy to share the data extracted from each included paper.

Acknowledgments: This manuscript builds on a Ph.D. thesis, and we would like to acknowledge the Australian Government Research Training program and Flinders University for supporting the study through a student support payment and fieldwork funding. We would also like to acknowledge Flinders University librarian−Shannon Brown for assisting the searching process of this systematic review.

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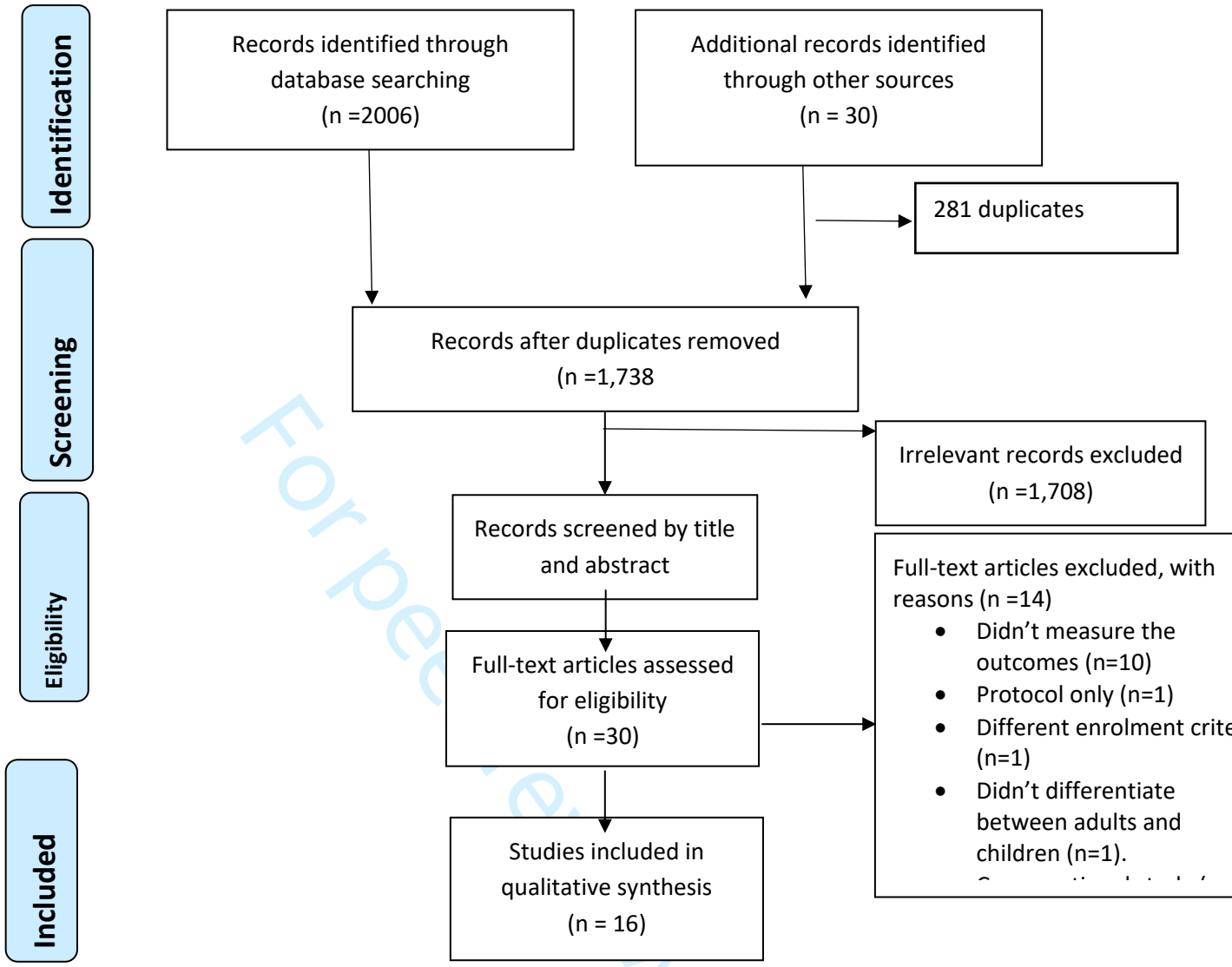


Figure 1: PRISMA flow chart

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Supplementary file 1: Shows the searching terms and outputs for Medline

#	Searches	Results
1	HIV Infections/ or HIV/ or HIV-2/ or HIV-1/	229102
2	Acquired Immunodeficiency Syndrome/	75776
3	(HIV* or AIDS* or "Acquired Immunodeficiency Syndrome").tw,kf.	382368
4	1 or 2 or 3	420727
5	Nutritional Status/	41768
6	Dietary Supplements/	52953
7	Food Assistance/ or *Food/ or *"Diet, Food, and Nutrition"/ or Food, Fortified/	28061
8	nutrition therapy/ or diet therapy/ or nutritional support/	18224
9	((food or nutrition) adj3 (support or program or intervention or supplement* or therap* or care or consult*)).tw,kf.	24448
10	5 or 6 or 7 or 8 or 9	151790
11	"africa south of the sahara"/ or africa, central/ or cameroon/ or central african republic/ or chad/ or congo/ or "democratic republic of the congo"/ or equatorial guinea/ or gabon/ or "sao tome and principe"/ or africa, eastern/ or burundi/ or djibouti/ or eritrea/ or ethiopia/ or kenya/ or rwanda/ or somalia/ or south sudan/ or sudan/ or tanzania/ or uganda/ or africa, southern/ or angola/ or botswana/ or lesotho/ or malawi/ or mozambique/ or namibia/ or south africa/ or swaziland/ or zambia/ or zimbabwe/ or africa, western/ or benin/ or burkina faso/ or cabo verde/ or cote d'ivoire/ or gambia/ or ghana/ or guinea/ or guinea-bissau/ or liberia/ or mali/ or mauritania/ or niger/ or nigeria/ or senegal/ or sierra leone/ or togo/	197050
12	("africa south of the sahara" or africa*or cameroon or central african republic or chad or congo or "democratic republic of the congo" or equatorial guinea or gabon or "sao tome and principe" or burundi or djibouti or eritrea or ethiopia or kenya or rwanda or somalia or south sudan or sudan or tanzania or uganda or angola or botswana or lesotho or malawi or mozambique or namibia or "south aftrica" or swaziland or zambia or zimbabwe or benin or burkina faso or "cabo verde or cote d'ivoire" or gambia or ghana or guinea or guinea-bissau or liberia or mali or mauritania or niger or nigeria or senegal or sierra leone or togo).mp.	361692
13	11 or 12	412824
14	4 and 10 and 13	616
15	limit 14 to english language	604
16	limit 15 to yr="2005 -Current"	510

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Supplementary file 2: JBI Critical Appraisal Checklist for Quasi-Experimental Studies (non-randomized experimental studies)

Reviewer_____Date_____

Author_____Year_____Record Number _____

	Yes	No	Unclear	Not applicable
1. Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	☐	☐	☐	☐
2. Were the participants included in any comparisons similar?	☐	☐	☐	☐
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	☐	☐	☐	☐
4. Was there a control group?	☐	☐	☐	☐
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	☐	☐	☐	☐
6. Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	☐	☐	☐	☐
7. Were the outcomes of participants included in any comparisons measured in the same way?	☐	☐	☐	☐
8. Were outcomes measured in a reliable way?	☐	☐	☐	☐
9. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

_____

_____

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Supplementary file 3: Assessment of methodological quality (n=16).

Authors	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	%
Ahoua, et al. (2011)	Y	N/A	N/A	Y	Y	Y	Y	Y	Y	100
Bahwere, et al. (2009).	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Berhe, et al. (2014)	Y	N/A	N/A	N	Y	Y	Y	Y	Y	86
Diouf, et al. (2016).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Mallewa, et al. (2018).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Manary, et al. (2010)	Y	Y	Y	Y	Y	Y	Y	Y	UC	100
Nagata, et al. (2014)	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Ndekha, et al. (2009)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Oguntibeju,et al. (2007)	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Olsen, et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Rawat, R., et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Rawat, et al. (2010).	Y	N/A	N/A	N	Y	Y	Y	Y	Y	86
Sadler, et al. (2012).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Scarcella, et al. (2011)	Y	Y	Y	N	Y	Y	Y	Y	Y	89
Tesfay, et al. (2020).	Y	Y	Y	N	Y	Y	Y	Y	Y	89
van Oosterhout, et al.	Y	Y	Y	Y	Y	Y	Y	Y	Y	100

Q= Question Y=Yes; N= No; NA= Not applicable, UC=unclear

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Annex table 1 Improvements in nutritional status and recovery and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food type	Improvement in nutritional status or recovery	Factors that affect nutritional recovery
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUTF	328 (62.2%) recovered. **Measurement**: BMI>=18.5 kg/m^2 in two subsequent measurements	The factors that were associated with nutritional recovery were: - female (adjusted odds ratio (AOR) and 95% confidence interval (CI)= 2.16 (1.340-3.473)) - educated (AOR and 95%= 1.82(1.110-2.983)) - moderate malnutrition (AOR and 95%= 2.594 (1.251-5.377)) - mild malnutrition (AOR and 95% =14.25 (7.074-28.742)) - WHO stage I and II (AOR and 95% CI =3.345 (1.652-6.772)) - no opportunistic infection during admission (AOR and 95% =2.791(1.047-7.439))
Mallewa, et al 2018 31, Africa	Randomized clinical trial	-Adults -At least five years with HIV infection and CD4 cell count fewer than -100	1805 -897 to treatment -908 to non RUSF	Peri-urban regional hospital **Food type**: RUTF and RUSF	Adults and adolescents aged 13 years and older in the RUSF group had significantly greater gains in BMI, and MUAC than the no-RUSF group	Not assessed

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		-Starting ART at the health facilities			(p=0·004, and 0·03, respectively). **Measurement:** Improvement in weight, BMI and MUAC (no cut off point)	
Nagata et al 2014 40, Kenya	Retrospective cohort study	-non pregnant adult patients -BMI less than 18.5 kg/m^2	1017	Rural village **Food type**: fortified corn-soy flour supplements	On average, enrolled clients gained 2.01 kg in weight and 0.73 kg/m^2 in BMI over follow-up (mean 100 days) with the highest gain among those severely malnourished. - Only 13.1% nutritional recovery according to criteria **Measurement**: BMI.20 kg/m^2	The factors associated with BMI >20 Kg/m^2 were - Greater BMI at baseline 1.52 (1.23–1.86) - younger age, 0.97 (0.94–1.00) - male gender, (0.51 (0.30–0.86)) - not requiring highly active antiretroviral therapy (HAART) (4.18 (2.42–7.22)

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Ndekha et al 2009 34, Malawi	Randomised control trial	- Adults living with HIV - BMI<18.5 Kg/m^2	491	Clinic based **Food type**: fortified corn-soy flour supplements	After 14 weeks, patients receiving fortified spread had a greater increase in BMI and fat-free body mass than those receiving corn-soy blend: 2.2 (SD 1.9) v 1.7 (SD 1.6). Measurement: Gain in BMI (no cut off point)	No significant differences in the CD4 count, HIV viral load, assessment of quality of life, or adherence to antiretroviral therapy were noted between the two groups
Oguntibeju et al 2007 41, South Africa	Retrospective cohort study	- non pregnant adults with HIV, non-diabetic and non-hypertensive - not on ART - CD4 cell count 100-350 cells/mm^3	35	Not specified **Food type**: extract of hypoxis, grape fruit seed extract, sitosterol and sitosterolin, beta-carotene, vitamin E, vitamin, vitamin B1, vitamin B2, vitamin B12, nicotinamide, vitamin C, olive green	19 (67.9%) of the 28 patients had a BMI within the normal range after nutrient intervention **Measurement:** BMI>=18.5kg/m^2	- significant positive correlation between the BMI and fat percentage (statistics not reported) - CD4+T cell count showed no correlation with any of the anthropometric indices - high viral load has significant negative correlation with the lean body mass and BMI (statistics not reported)

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				leaf extract, folic acid and natural antioxidant (biocydin)		
Olsen et al 2014 33, Ethiopia	Randomised control trial	- Age greater than 18 years - BMI greater than 16 Kg/m^2	453	Referral hospital **Food type**: Plump sup	At three months, participants receiving the supplements containing whey or soy had increased their lean body mass by 0.85 kg (95% confidence interval 0.16 kg to 1.53 kg) and 0.97 kg (0.29 kg to 1.64 kg), respectively, more than controls. **Measurement**: Weight gain and BMI (no cut off point)	- No effect on physical activity - The effects of the two supplements, however, were not significantly different in direct comparison
Rawat et al, 2014 35, Uganda	Prospective cohort	- Food insecure people living with HIV who are new to ART. - employment status and income	904	Community based **Food type**: WFP Rations	Over 12 months, food assistance significantly increased body mass index by 0.6 kg/m2 (P, 0.01) and mid-upper arm	- no impact on CD4 count, Hb concentrations, or Individual Dietary Diversity Score - at the household level, food assistance increased the Household Food Insecurity Access Scale, by 2.1 points (P, 0.01) - food assistance increased BMI by 0.6 kg/m2 and MUAC by 6.7 mm.

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		- housing characteristics and past experience of food insecurity			circumference by 6.7 mm (P, 0.05). **Measurement:** Change in BMI (no cut off points)	- no statistically significant impact of food assistance on Hb concentration or CD4 count
Sadler, et al, 25 Ethiopia	Comparative follow up study. Report, not peer reviewed	Adults living with HIV - BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 intervention were included in the final analysis	Health facilities / health centres **Food type**: RUTF/RUSF	636 (32.6) graduated in the intervention group and 120 (18.8) in the comparison group **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Factors that were associated with nutritional recovery were: - being in the intervention group 2.4 (1.6-3.4) - female 1.4 (1.0-1.9) - attending in a hospital 0.5 (0.4-0.7)
Scarcella. et al 43, Mozambique	Randomised control trial	- Adult HIV patients - BMI <18.5 Kg/m^2 - presence of wasting - TB co-infection - insufficient access to food - CD4 count < 200	106	Public Health Centre **Food type**: - food package consisted of a ration of cereals (rice or corn) of around 15kg; pulses (5kg); peanuts (3kg); sugar (2kg); sunflower seed oil (3Lt.);	In the cohort not in program the mean BMI values showed an increase but not significant (initial value: 21.9±2.9; final value: 22.5±3.3) **Measurement**: Increase in BMI (no cut off point)	- haemoglobin showed an increase but not significant - in the intervention group both, BMI and Haemoglobin shown a significant increase - increase in BMI was statistically associated with the dietary diversity score in the intervention group

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				at least 7kg of fortified corn soy blend		
Tesfay et al, 42 Ethiopia	Retrospective cohort study	- Adults living with HIV - BMI<18.5	1757	Hospitals **Food type**: RUTF/RUSF	55.3% (95% CI = 53.2–57.4) recovered from malnutrition **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Not assessed
van Oosterhout et al 2010 37, Malawi	Retrospective cohort study	-Adults aged 18 years and older -BMI 18.5 kg/m^2 -ART naïve or - WHO clinical stage III or IV Or - CD4 count B250 cells/ml irrespective of clinical stage	593	Centeral hospital **Food type**: corn/soy blended flour (CSB) or ready-to-use fortified spread (RUFS), a peanut butter-based product	- Linear regression analysis showed that increase in BMI was greatest in the RUSF group and better in the CSB group than in those receiving no food supplementation at 14 weeks. - These differences were no longer significant at 26 weeks **Measurement:** Increase in BMI	- no longer a significant effect of supplementary feeding on an increase in BMI, 12 weeks after the feeding was stopped. - factors that were associated with nutritional recovery or improved BMI was not assessed.

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Annex table 2: Shows program failure (default + nonresponse +death) after enrolment in a nutritional program in sub Saharan Africa

Author, year and country	Study design	Study population	N	Setting and type of food	Nutritional program failure	Factors that affect nutritional program failure
Ahou et al 2011 38, Kenya and uganda	Retrospective cohort study	Adults aged 15 years and older living with HIV BMI less than 17 kg/m^2 and bilateral pitting oedema	1106	Regional hospital Type of food: RUTF/RUSF	An overall total of 531 of (48.0%) patients failed nutrition therapy	Factors that were associated with program failure were: - male gender (OR = 1.5, 95% CI 1.2-2.0), - patients with severe malnutrition at nutrition programme enrolment (OR = 2.2, 95% CI 1.7-2.8), and - those never started on ART (OR = 4.5, 95% CI 2.7-7.7) for those eligible; OR = 1.6, 95% CI 1.0-2.5 for those ineligible for ART at enrolment
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital Type of food: RUTF/RUSF	Overall program failure rate was 196 (37.6%)	The factors that were associated with nutritional program failure were not assessed.

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Annex table 3: Nonresponse to the nutritional program and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food	Nonresponse to the nutritional program	Factors that contribute to non-response to the nutritional program
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUFT/RUSF	- 155 (29.6%) non-respondent - **Measurement**: not reaching 18.5 kg/m^2	The factors that affect non-response were not assessed
Sadler, et al , 25 Ethiopia	Comparative follow up study. not peer reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each. 1956 and 639 were including in the final analysis	Health facilities / health centres **Food type**: RUFT/RUSF	- 1316 (67.4%) non-respondent in the intervention group and 519 (81.2) become non-respondent in the comparison group - **Measurement**: not reaching 18.5 kg/m^2	Factors that were associated with nutritional non-response were: - attending the program at a hospital was associated with an increased likelihood of non-response (OR=1.75, p=0.021), - severe food insecurity was associated with a decreased likelihood of non-response (OR=0.46, p=0.016) compared to those who were "food secure.
Tesfay et al , 42 Ethiopia	Retrospective cohort study	Adults living with HIV and-BMI<18.5	1757	Hospitals **Food type**: RUFT/RUSF	- 21% (95% CI = 19.7–23.4) completed the program but failed to recover from malnutrition. - **Measurement**: not reaching 18.5 kg/m^2	- urban residence, (AOR = 1.46, 95% CI = 1.4–2.91), - attending Shul (AOR = 2.92, 95% CI = 2.04–4.19) and - Lemlem Karl (AOR = 1.49, 95% CI 1.05–2.11) hospitals

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Annex table 4: Default from the nutritional program and its determinants in sub-Saharan Africa

Author, year, and country	Study design	Study population	N	Setting	Default to the nutritional program	Factors that contribute to default to the nutritional program
Sadler, 25 et al , Ethiopia	Comparative follows up study. not peer-reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 intervention were included in the final analysis	Health facilities / health centres	70.6% defaulted from the nutritional program among the intervention group	- severe acute malnutrition (OR: 1.89, P-value=0.003)
Tesfay et al 42 , Ethiopia	Retrospective cohort study	Adults living with HIV and BMI<18.5	1757	Hospitals	19% (95% CI, 17.3–20.7) did not complete the program.	Factors associated with program incompleteness. - living in urban areas (AOR = 1.44, 95% CI = 1.05– 1.97), - employed (AOR = 1.39, 95% CI = 1.01–1.93), - attending Shul (AOR = 4.6, 95% CI = 3.15–6.71) and Lemlem Karl (AOR = 2.5, 95% CI = 1.69–3.71) hospitals, - in clinical stages II (AOR = 2.49, 95% CI = 1.59–3.91) and III (AOR = 1.46(1.02–2.07), - on ART for less than six months (AOR = 1.61, 95% CI = 1.09–2.39), - anaemic (AOR = 1.77, 95% = 1.29–2.41), and - severe acute undernutrition at enrolment (AOR = 6.43, 95% CI = 4.69–8.3)

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	N/A

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	N/A
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	26
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	30
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	30
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	No funding

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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BMJ Open

Characteristics and impacts of nutritional programs to address undernutrition of adults living with HIV in sub-Saharan Africa: A systematic review of evidence.

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-047205.R2
Article Type:	Original research
Date Submitted by the Author:	04-Nov-2021
Complete List of Authors:	Tesfay, Fisaha; Mekelle University College of Health Sciences, School of Public Health Javanparast, Sara; Flinders University, College of Medicine and Public Health Gesesew, Hailay ; Flinders University, College of Medicine and Public Health Mwanri, Lillian; Flinders University, College of Medicine and Public Health Ziersch, Anna; Flinders University, Southgate Institute for Health, Society and Equity
Primary Subject Heading:	HIV/AIDS
Secondary Subject Heading:	Epidemiology, Health services research, HIV/AIDS
Keywords:	PUBLIC HEALTH, NUTRITION & DIETETICS, EPIDEMIOLOGY, Infectious disease/HIV < NEUROLOGY

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Characteristics and impacts of nutritional programs to address undernutrition of adults living with HIV in sub-Saharan Africa: A systematic review of evidence.

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Abstract

Objectives: Although some studies have identified various challenges affecting nutritional programs to effectively tackle undernutrition amongst people living with HIV, evidence about the characteristics and impacts of these programs on weight-related nutritional outcomes varies based on country contexts, specific program goals and the implementation processes. This systematic review sought to synthesise evidence on the characteristics and impact of nutritional programs on weight-related nutritional outcomes of people living with HIV in sub-Saharan Africa.

Design: Systematic review

Data sources: We searched for primary studies published in the following databases: Web of Science, Medline, Scopus, ScienceDirect, ProQuest and Google Scholar, supplemented by checking reference lists of included papers.

Eligibility criteria: Studies published from 2005 to the 10th of July 2020 and reporting on the weight-related nutritional outcomes of undernourished people enrolled in nutritional programs in HIV care in Sub-Saharan Africa were included.

Data extraction and synthesis: Data were extracted using a data extraction proforma. Weight-related nutritional outcomes of people living with HIV before and after enrolment in a nutritional program were compared and narratively synthesised.

Results: Sixteen studies assessing the impact of nutritional programs in HIV care on weight-related nutritional outcomes were included**.** Of these, 13 examined nutritional programs implemented in health facilities and the remaining three were delivered outside of health facilities. Nutritional recovery (defined differently in the studies) ranged from 13.1% to 67.9%. Overall program failure rate, which included default after enrolment in a nutritional program or non-response, ranged from 37.6% to 48.0%. More specifically, non-response to a nutritional program ranged from 21.0% to 67.4% and default from the program ranged from 19.0% to 70.6%. Key sociodemographic, clinical, and nutritional characteristics that affect nutritional recovery, non-response and default were also identified.

Conclusions and recommendations: Nutritional programs in HIV care have led to some improvements in weight-related nutritional outcomes amongst people living with HIV. However, the programs were characterised by a high magnitude of default and non-response.

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To improve desired weight-related nutritional outcomes of people living with HIV, a holistic approach that addresses longer-term determinants of undernutrition is needed.

Keywords: Nutritional program, nutritional support, HIV and nutrition

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Strengths and limitations

- This review is the first of its kind to synthesise the evidence with regards to weight-related outcomes of nutritional programs to address undernutrition for people with HIV across sub-Saharan Africa, which has the largest number of people living with HIV.
- The review assesses multiple weight-related nutritional outcomes and presents a summary of evidence from a range of nutritional programs.
- However, the diversity and heterogeneity of the studies and nutritional programs meant that meta-analysis was not possible to be done, a shortcoming of the review.
- This study only focused on papers published in English which may have excluded potential studies on the same topic but published in other languages.

The original protocol for the study: The study protocol is attached at the end of the study.

Funding statement: Not funded.

A competing interest statement. All authors declare that they have no conflict of interest.

Introduction

Evidence demonstrates the clinical and social susceptibility of people living with HIV to undernutrition and its wider effects [1] [2]. Inadequate food intake due to lack of appetite, malabsorption, and impaired storage and metabolism are clinical causes of undernutrition among people living with HIV [3], while social factors such as poverty, gender inequality, food insecurity, stigma, and discrimination also hinder access to adequate food and nutrition [4] [5]. Undernutrition involves weight-related stunting (low height for age), wasting (low weight for height), underweight (low weight for age) and micronutrient deficiencies but this particular review will focus on undernutrition [6]. Weight related undernutrition (hereafter undernutrition) has various impacts on the wellbeing of people living with HIV, including on HIV-related mortality in resource poor settings which has been highlighted in a range of studies [7-9]. Research has also indicated that undernutrition can act as a barrier to the utilisation of HIV care services and antiretroviral therapy (ART) adherence [10-12].

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The World Health Organisation (WHO) highly recommends the integration of nutritional interventions into HIV care and support services [13] [14]. To address the impact of undernutrition among people living with HIV, nutritional programs targeting people living with HIV are operational in resource-poor settings, mainly in sub-Saharan Africa [15]. The key elements of these programs are nutritional assessment, counselling, and nutritional support [16-18]. The latter component includes the provision of micronutrient, macronutrient, or fortified macronutrient supplementation [19] [20].

Concept 1: ***Nutritional program***	**Concept 2:** ***HIV/AIDS***	**Concept 3:** ***Sub-Saharan Africa***
Nutritional program	HIV-1	South of Saharan Africa

Nearly 90% of HIV care sites in sub-Saharan Africa have responded to the WHO recommendations and instituted some forms of nutritional program [21], with substantial variations between and within countries [17] [19] [22]. Donor organisation policies regarding the nutritional program and availability of national and local resources have led to program differences observed between countries in sub-Saharan Africa [23], including the type of care [24-26] and nature of nutritional support provided [27-29]. Evidence regarding the impact of nutritional programs on nutritional outcomes is fragmented, mixed, and varies by country context, program goals, and implementation. This study aims to systematically review and synthesise evidence on the characteristics and impact of nutritional programs on nutritional outcomes of undernourished people enrolled in nutritional programs in HIV care in sub-Saharan Africa.

Methods

Protocol

The protocol for this systematic review has been registered under the PROSPERO review registration number: CRD42020196827.

Systematic search strategy

This review was guided by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). Studies published in English language in the Web of Science, Medline, Scopus, ScienceDirect, ProQuest, from 2005 to the 10th of July 2020 were searched. Google Scholar and the reference lists of references of selected papers were also searched. The search concepts and terms included “Nutritional programs”, “HIV/AIDS”, and “sub-Saharan Africa” (**Table 1**). Outputs of Ovid Medline are shown in the **supplementary file 1**. **Table 1:** Concepts and their respective keywords

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Nutritional intervention	HIV-2	Africa
Nutritional care	HIV infection	Western Africa
Food by prescription	Human immunodeficiency syndrome	Central Africa
Supplementary food	Acquired immune deficiency syndrome	
Ready to use Therapeutic food		
Food supplementation	AIDS	List of all countries in sub-Saharan Africa

Selection of studies

The screening process involved two stages. The first stage was conducted by FT and involved screening the studies by title and abstracts against the inclusion criteria. The second stage was conducted by two independent reviewers (FT and HAG) and involved full-text screening of articles. The inclusion criteria were:

- longitudinal studies (cohort and interventional) that examined nutritional outcomes of undernourished people (age>15 years) who were enrolled in nutritional programs in HIV care settings.
- studies measuring nutritional outcomes using Body Mass Index, (BMI), weight gain or Mid Upper Arm Circumference (MUAC)
- studies published since 2005, because most nutritional programs in sub-Saharan Africa started in 2005 and published in English.

Studies involving children and pregnant women were excluded because different enrolment and nutritional outcome criteria are used to enrol children and pregnant women in programs. Studies were also excluded from the review if information about age or pregnancy status was not provided (**Figure 1**).

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Measurement of exposures

The exposure variables included:

- sociodemographic factors: age, sex, residence, marital status, and educational status.
- clinical factors: WHO clinical stage, CD4 count, functional status, presence of opportunistic infections, and HIV status disclosure.
- nutritional factors: baseline nutritional status, the status of food security and other dietary information.

Measurement of outcomes

The outcome variables were nutritional outcomes of undernourished adults living with HIV, compared before and after enrolment in a nutritional program. Outcomes included changes in BMI, weight, or MUAC and nutritional recovery (BMI$\geq$ 18.5 kg/m^2), default, or non-response to the nutritional program.

Context/setting

Most nutritional programs in sub-Saharan Africa commenced in 2005. Most nutritional programs in sub-Saharan Africa included in the reviewed studies had similar program components which included nutritional assessment, counselling, and nutritional support. This review is limited to the sub-Saharan Africa given the duration, enrolment, and exit criteria of nutritional programs are very different in other regions and settings.

Assessment of methodological quality

The Joanna Briggs Institute 30 for critical appraisal tool for quasi-experimental studies was used to assess the quality of studies that were included in the review. Two independent reviewers (FT and HAG) conducted the quality assessment.

Data extraction

Data were extracted using a proforma adopted from JBI-MAStARI. The extracted data involved information about the outcome and exposure that affected the nutritional outcomes. In addition, population, sample size, country description of the nutritional program, and the conclusion of the studies were extracted.

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Data analysis

Due to the heterogeneity of the studies included in terms of study design, the measure of nutritional outcomes used, and differences in enrolment and exit criteria, a narrative synthesis was the most appropriate analysis procedure for this review. Different variables were identified from the reported findings of each study. Similar patterns were grouped in subsequent coding and analysis performed. Finally, the findings were categorised into two major themes.

Patient and Public Involvement: No patients or public was involved in this study. This is a systematic review and it is not applicable to make patient and public involvement.

Results

Description of the studies

A total of 2036 papers were obtained from electronic databases and other sources. After removing duplicates 1,738 studies were screened by title and abstract and 1,708 were deemed irrelevant. The remaining 30 papers were scrutinised further against the inclusion criteria using full-text screening. Sixteen papers involving 25,335 participants were included in the final review (**Figure 1**). Of the 16 final studies included in the review, nine were retrospective cohort studies, three were prospective cohort studies, and four were randomised trials.

Assessment of methodological quality

The methodological and reporting quality of the reviewed studies was assessed using the JBI critical appraisal tool for quasi-experimental studies (**supplementary file 2**). The assessment tool included nine parameters each carrying equivalent weight (**supplementary file 3**).

Nine studies, six clinical trials 31-36, and three cohort studies 26 37 38 scored nine out of nine (100%).

One retrospective cohort study 39 scored seven out of nine but two parameters (related to the comparability of study participants and similarity of interventions in both the exposed and non-exposed groups) were not applicable− scoring 100%. Three cohort studies scored six out of nine which is 100% as the same two parameters were not applicable, and the presence of control groups was not clear 40-42.

Two retrospective studies scored eight out of nine (89%) as there was no control group 43 44. Another retrospective cohort study scored six out of nine, with no control group and two parameters not applicable - a final score of 86% 45.

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Characteristics of the nutritional programs

The nutritional programs examined in this review varied in terms of enrolment criteria, duration, and type of nutritional support used. Six studies used BMI <18.5 kg/m^2 as an enrolment criteria 26 34 35 41 43 45. Another seven studies used one of the following: bilateral oedema, being bedridden for 50% of the month, TB coinfection, CD4 count < 350, WHO clinical stage III and IV, in addition to BMI< 18.5 kg/m^2 or MUAC <210mm) 32 33 38-40 42 44. Living in a household with HIV positive family members was used as an enrolment criterion in one study 37. Another study enrolled people living with HIV at any stage of the disease and not able to eat 31. A further study used the world food program (WFP) poverty assessment index, not receipt of other food assistance, CD4 count between 200 and 450 cells per microliter, and ART naïve (new for the treatment) as an enrolment criterion 36.

In five studies, the duration of the nutritional program ranged from three months (for mild and moderate undernutrition) to six months (for severe undernutrition) 26 39 40 43 45. Two studies reported nine weeks of program duration 33 44, two had 14 weeks 33 35, and a further study 32 24 weeks duration before individual patients exited from the program. The nutritional program was nine months in one study 38 and twelve months in three studies 34 36 37. No duration limit was reported in two studies 41 44.

Various types of nutritional supports were used in nutritional programs. Ready to use therapeutic food (RUTF) or ready to use supplementary food (RUSF) were used in eight studies 26 31 32 34 39 40 43-45. Corn/soy blended food was used in three studies 33 35 38 41 and the WFP food basket (which involves generic food rations) was used in two studies 36 37. One study used an extract of hypoxis, grapefruit seed extract, sitosterol and sitosterolin, beta-carotene, vitamin E, vitamin, vitamin B1, vitamin B2, vitamin B12, nicotinamide, vitamin C, olive green leaf extract, folic acid, and natural antioxidant (biocydin) as the nutritional supplement 42.

Although nutritional assessment was an element of nutritional programs in all studies included in this review, the provision of nutritional counselling was stated only in two studies 34, 43

The nutritional programs were provided in health facilities such as hospitals and clinics in 12 studies 26 31-35 37-39 43-45, in community-based settings in four studies 34 40 41, and no setting was specified in one study 42.

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Impact of the nutritional program on weight related nutritional outcomes (nutritional outcomes here after)

In reporting improvements in the nutritional outcome of people living with HIV, studies used various outcome measures. Gain in BMI, weight, and MUAC were used either in combination or as separate nutritional outcome measures. BMI was used in nine studies [26] [35] [36] [38] [39] [42-45] and both MUAC and BMI were used in two studies [32] [40]. The other two studies used weight gain [31] [37], and three studies used both BMI and weight gain to measure the nutritional outcomes [31] [34] [37].

Nutritional program failure, recovery, default, and non-response were main outcomes measured in most review studies. The definition of these terms in the studies included are shown in **table 2.**

Table 2: The definition of nutritional outcomes in nutritional programs

Outcomes	**Adults**
Program failure	Includes default and non-response from the nutritional program
Improvement in nutritional status and recovery /graduation from the nutritional program	Various criteria were used to determine nutritional recovery • Reaching BMI of 18.5kg/m^2 for two consecutive visits in some studies or BMI>18 kg/m^2 or BMI>20kg/m^2 In other studies, the nutritional outcome was measured in terms of improving weight, BMI, or MUAC.
Non-response	Participants did not reach the target BMI for two consecutive visits or did not show any change in BMI, weight, and MUAC
Default /failure to complete the nutritional program	Dropped out of the program before reaching the target BMI, weight or MUAC

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Improvement in nutritional status or recovery from the nutritional program

Eleven studies measured nutritional recovery which was defined differently in various nutritional programs. Three studies defined program recovery as BMI>18.5kg/m^2 in two consecutive measurements 26 43 45, and nutritional recovery in these studies ranged from 32.6% 26 to 62.2% 45. One study used BMI >18 kg/m^2 in two subsequent measurements and the absence of oedema, and recovery was 47.4% 39. A further study defined program recovery as BMI>20kg/m^2 and 13.1% attained nutritional recovery against these criteria 41.

Four studies used improvements in BMI, without specifying a specific cut-off point 35 36 38 44. The improvement in BMI in these studies ranged from 0.6 kg/m^2 using WFP rations for 12 months 36 to 2.2 kg/m^2 in a study where participants received a fortified spread for 14 weeks 35 and those who used a corn-soy blend showed a BMI increase of 1.7 kg/m^2 35. Weight gain was used to measure nutritional outcomes in four studies 31 34 37 40 and weight gain ranged from 0.85 kg 34 to 3 kg 40.

One study using weight gain found that after 9 weeks of supplementation, body weight, and fat-free mass increased significantly by 11 % (p= 0.033), and 11.8 % (p= 0.033) in the RUTF group, but not in the control group 31. One study used BMI, weight gain, and MUAC 32, and those in the RUSF group had significantly greater gains in weight, BMI, and MUAC than those in the no-RUSF group (p=0·004, and 0·03, respectively).

Annex table 1 presents the socioeconomic, demographic, clinical, and nutritional determinants of nutritional outcomes among those enrolled in nutritional program. For sociodemographic factors that affected nutritional outcomes, female gender was associated with nutritional recovery in three studies 45 26 41. Compared with those who didn't attend any schooling, individuals who attended school were also more likely to recover than those who did not in one study 45, and attending the nutritional program in a hospital was inversely related to nutritional recovery in another study 26.

Few clinical characteristics were associated with nutritional recovery. In one study, not being on ART was associated with nutritional recovery 41. In another study, compared with WHO clinical stage 4, being at WHO clinical stage 1 and 2 were associated with nutritional recovery 45. In addition, the absence of opportunistic infection at enrolment to the nutritional program was associated with nutritional recovery compared with those who had an opportunistic infection 45.

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Regarding the impact of nutritional characteristics at enrolment, compared with those enrolled with severe undernutrition, those with moderate undernutrition were associated with nutritional recovery in two studies 41 45. In one study, enrolment in the nutritional program with mild undernutrition was associated with nutritional recovery compared with those with severe undernutrition 45.

Program failure rate

The program failure rate involves a combination of default, non-response, and mortality while enrolled in the nutritional program. The program failure rate was reported in two studies 39 45. The nutritional program failure rate in a study from Kenya and Uganda was 48% 39 and 37.6% in a study from Ethiopia 45. Factors that were associated with the program failure were only assessed in one study 39 which found male gender, severe undernutrition at program enrolment (pre-ART), and ineligibility for ART at enrolment as variables associated with program failure (**Annex table 2**).

Non-response from the nutritional program

Three studies from Ethiopia measured non-response to the nutritional program ranging from 21% 43, to 29.6% 45, and 67.4% 26.

Compared with those who attended primary health care services to receive nutritional services, attending a hospital was associated with greater non-response in one study 26. A study by Tesfay and colleagues examining health facility based nutritional programs in Ethiopia reported that being an urban area was associated with non-response 43.

In terms of clinical characteristics, the same study found advanced WHO clinical stage (stage IV), and bedridden functional status were associated with non-response to the nutritional program 43. Compared with those enrolled with mild undernutrition, severe undernutrition at enrolment was also associated with non-response 43.

In another study, those with severe food insecurity were less likely to not respond than those who were food secure 26. The third study did not assess factors contributing to non-response (**Annex table 3**).

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Default from the nutritional program

Two studies from Ethiopia reported program default and associated factors.

In one study, 19% of the study participants did not complete the nutritional program. Factors that were associated with default were living in urban areas, being employed, attending at a hospital, and in clinical stages II and III 43. In another study, the default rate was 70.6% 26. Being on ART for less than six months and being anaemic were associated with default 26 (**Annex table 4**).

In both studies, severe undernutrition at enrolment to the nutritional program was associated with default from the nutritional program 26 43.

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Discussion

Given the susceptibility of people living with HIV to undernutrition, and the synergistic effect of undernutrition in accelerating HIV infection to AIDS, scaling up nutritional programs in HIV care in combination with other HIV services is vital 46. Despite the provision of nutritional programs in HIV care settings in many countries 22, evidence regarding the effectiveness of these programs is fragmented–this review sought to address this with regards to sub-Saharan Africa.

Broadly, a range of anthropometric, clinical, and socioeconomic parameters were used as enrolment criteria in nutritional programs in the HIV care setting in sub-Saharan Africa. The duration of the nutritional programs was characterised by a significant variation between programs. For instance, nutritional programs that aimed at the treatment of acute undernutrition were characterised by a shorter duration (ranging from three to six months) and provision of highly dense nutritional support (RUTF or RUSF) 26 39 43. However, there were examples where nutritional support was provided for more than nine months 32 33 35. The duration of nutritional programs that used corn-soy blended ranged from 14 weeks 33 35 to 9 months 38, and 12 weeks in programs that used the WFP food rations 36.

There was no optimal duration of the nutritional program because of the difference in content and nutritional value of the nutritional support used from program to program. Nutritional programs that use dense nutritional supports such as RUTF and RUSF may achieve faster weight gain and perhaps contribute to a shorter duration in the nutritional program 26 43. These findings are consistent with outcomes from other studies conducted in other parts of the world such as India 47.

There is strong evidence in the literature regarding the role of nutritional programs in enhancing weight gain among those enrolled in a nutritional program with BMI<18.5 kg/m^2 48. However, in nutritional programs with predetermined program exit criteria, there was substantial variation in the magnitude of nutritional recovery. Nutritional recovery beyond attaining weight gain can vary from program to program depending on the availability of resources. In nutritional programs that used BMI>18.5kg/m^2 in Ethiopia, nutritional recovery was 32.6% 26, 55.3% 43 and 62.2% 45. Other studies did not use a defined cut-off point or used different BMI measure (e.g., one used BMI>18 and another used BMI>20) or other measures (weight gain, MUAC) to determine nutritional outcomes. All of these demonstrated that nutritional support of various forms were associated with better nutritional outcomes.

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A systematic review that examined the impact of nutritional supplementation on HIV nutritional and clinical prognosis in sub-Saharan Africa found increased weight gain 49.

Evidence from the current review showed that few studies reported on the rate of overall nutritional program failure. Two studies from Ethiopia found program failure rates involving non-response, default, and mortality rates that ranged from 37.6% 45 to 48.0%39.

Studies examining non-response to the nutritional program (not reaching a BMI of 18.5kg/m^2 in two consecutive visits) were all from Ethiopia. While the nutritional program was similar in terms of the duration, type of food supplement, funding organisation, and policy, there was a big variation in the magnitude of non-response rate between these studies of 21% 43, 29.6% 45, and 67.4% 26. The two studies 43 45 with a relatively low non-response rate were conducted recently, after several years of implementation of the nutritional program in Ethiopia, which may suggest that program non-response rates may have dropped due to the accumulation of experience by the program.

Two studies from Ethiopia measured default from the nutritional program and the difference in default between the two studies was substantial −ranging from 19% 43 to 70.6 % 26. While the study by Tesfay et al (2020) was conducted recently, the later study by Sadler (2012) was conducted at the initial stage of the nutritional program as a pilot evaluation and may reflect a less developed version of the program. Reasons for nutritional program ineffectiveness could include poor taste of the supplement, dietary boredom, or HIV-related complications such as thrush as highlighted in a systematic review that involved resource adequate and constrained settings 50.

It is very difficult to compare nutritional outcomes by country because of the lack of uniformity in the nature of the nutritional program in terms of the duration, enrolment criteria, and the nutritional support that is given. For instance, nutritional programs in Ethiopia used ready to use therapeutic or supplementary food for three months for mild and moderately undernourished and six months for severe undernutrition 26 43 45. On the other hand, nutritional programs from Kenya used ready to use therapeutic or supplementary foods provided in a rural village but no specific duration was given for the nutritional program 41. Nutritional recovery (achieving BMI greater than 20 kg/m^2) in Kenya 41 was lesser (13%) than studies from Ethiopia which were 62.2% 45, and 32.6% 26.

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The studies provided some insights into factors associated with program outcomes, including socio-demographic factors. Gender was highlighted, where being male was associated with nutritional program failure 39 and lower rates of recovery 41 26 45 and living in urban areas was associated with program non-response 43. In one study, attending the nutritional program in a hospital was associated with lower rates of nutritional recovery 26. Another study found differences between hospitals where attending the nutritional program at relatively rural hospitals was associated with higher rates of non-response 43.

Clinical factors were also linked to program outcomes. For example, not starting ART was associated with nutritional program failure in one study 39, but another study found it was associated with nutritional program recovery 41. This might be due to the fact that many individuals in the study by Ahoua et al 2011 were eligible for ART despite the lack of ART initiation because of delayed patient laboratory results or patient refusal. These patients might have died, giving rise to a high magnitude of program failure in the former study. Being on ART for less than six months were associated with default from the nutritional program 43. The absence of opportunistic infection at enrolment was associated with nutritional recovery in one study 39.

Severe undernutrition at baseline was associated with a higher nutritional program failure rate 39, lower likelihood of achieving nutritional recovery 41, and default from the nutritional program 43. On the other hand, mild and moderate undernutrition at enrolment was associated with nutritional recovery in one study 45. Compared with those who are not enrolled in the nutritional intervention, those enrolled in the nutritional program were more likely to recover 26. One study found that those with severe food insecurity were less likely to become non-respondents to the nutritional program 26.

Given the clinical, social, and economic susceptibility of people living with HIV to undernutrition 51, findings from the current review highlighted that nutritional programs in sub-Saharan Africa have focused on the treatment of undernutrition among people living with HIV 26 39 41. For instance, terms such as nutritional recovery, weight gain, default, and nutritional recovery were used in all of the studies included to imply the short term positive impact of the nutritional programs among people living with HIV. Measures used in all nutritional programs were clinical and none provided measures on longer-term nutritional wellbeing. Longer-term measures such as food security were not included as key program goals in any of the nutritional programs despite the wider vulnerability of people living with HIV to food insecurity in sub-Saharan Africa 52 53.

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Strength and Limitations

The key strength of this study is its coverage of sub-Saharan Africa, a major area of HIV infection, and its review of multiple nutrition programs representing a large population base. It also examines a range of nutrition outcomes. However, the review has some limitations., Some studies had a very small sample size (e.g. 42 (n=35) and 44 (n=106)) which may negatively impact the estimates of the nutritional outcomes and factors associated with them. It was not possible to pool estimates of nutritional outcomes because of the differences in enrolment and exit criteria and program duration between nutritional programs. This study only included papers published in English which automatically excluded papers published in other languages, which may lead to reporting bias. The focus of this review is on the impact of nutritional programs on undernourished people enrolled in nutritional programs in HIV care settings. However, overnutrition has also been highlighted as an important issue for people living with HIV as it increases the risk of overweight and cardiovascular disease among people with HIV on long-term ART treatment 54 55. An examination of programs designed to address this would be an important area for future reviews.

Conclusion and recommendations

Despite the use of various enrolment and exit criteria, nutritional programs in HIV care have led to overall improvements in nutritional outcomes and overall wellbeing of people living with HIV. However, the programs reviewed were characterised by problems of effectiveness such as a high magnitude of default and non-response. Various sociodemographic, clinical, and nutritional characteristics have an impact on the nutritional outcomes, which gives potential avenues for future research and program improvement. However, the goal of most nutritional programs in HIV care settings is to reverse short-term nutritional problems such as undernutrition. In contrast, the impact of nutritional programs in HIV care should be measured in terms of the program's capacity to sustainably address the nutritional wellbeing of individual HIV patients and others around them over the longer term 56.

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Author contributions: **FT** conceptualised, designed the study, conducted the search, analysed, and interpreted the data as well-drafted the manuscript. **HAG** assisted with the analysis. **SJ, AZ, HAG, and LM** critically reviewed the design, analysis, and interpretation of the findings and critically reviewed and approved the manuscript.

Any checklist and flow diagram: The PRISMA diagram was used.

A patient consent form: All primary studies were assessed for the fulfillment of ethical standards, and all fulfilled the ethical standards.

Data statement: The authors are happy to share the data extracted from the included papers.

Supplementary and raw data: We do not have supplementary raw data, but we are happy to share the data extracted from each included paper.

Acknowledgments: This manuscript builds on a Ph.D. thesis, and we would like to acknowledge the Australian Government Research Training program and Flinders University for supporting the study through a student support payment and fieldwork funding. We would also like to acknowledge Flinders University librarian−Shannon Brown for assisting the searching process of this systematic review.

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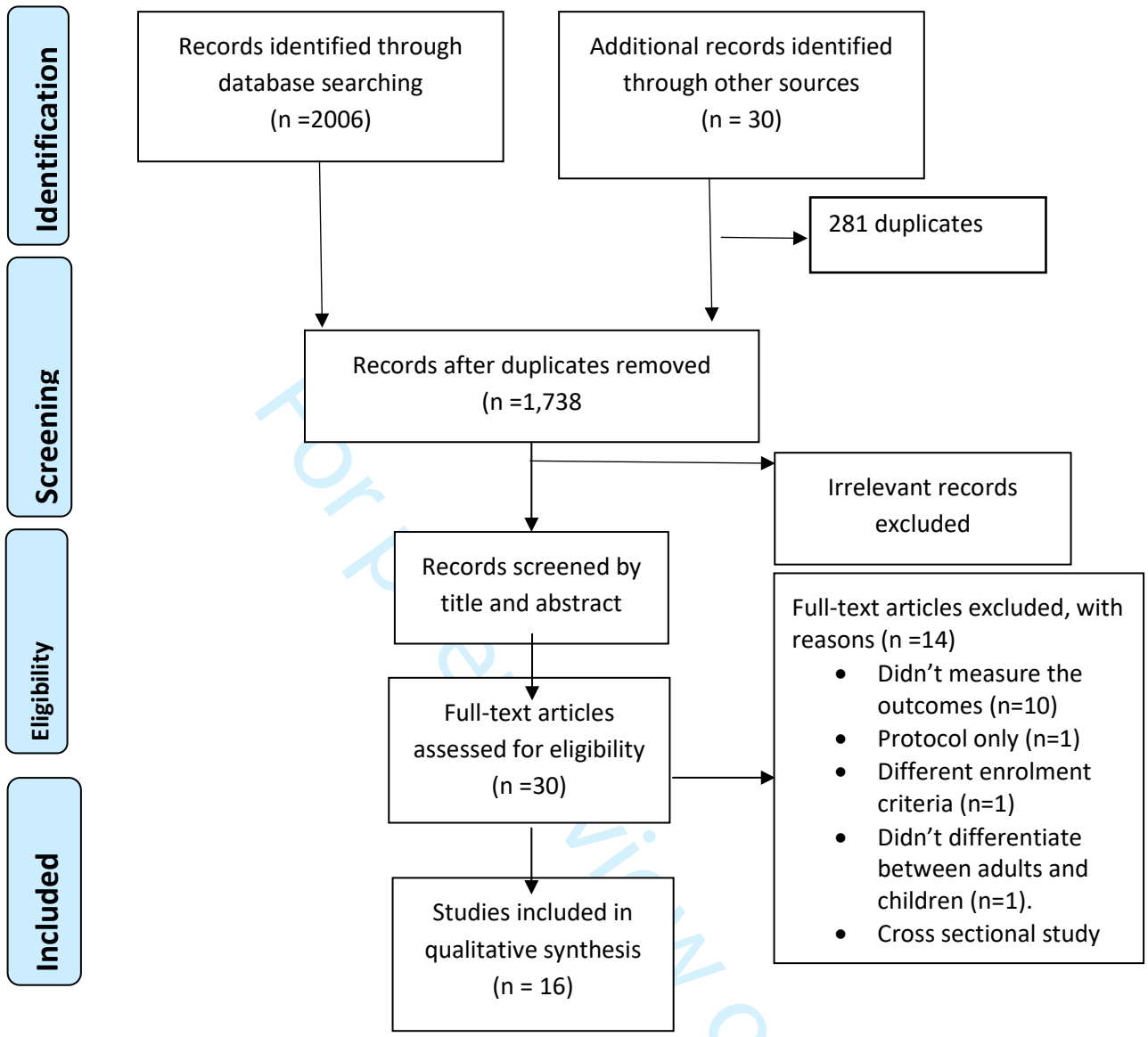


Figure 1: Flow diagram of literature search, selection and reasons for exclusion of removed studies

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Annex table 1 Improvements in nutritional status and recovery and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food type	Improvement in nutritional status or recovery	Factors that affect nutritional recovery
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUTF	328 (62.2%) recovered. **Measurement**: BMI>=18.5 kg/m^2 in two subsequent measurements	The factors that were associated with nutritional recovery were: - female (adjusted odds ratio (AOR) and 95% confidence interval (CI)= 2.16 (1.340-3.473)) - educated (AOR and 95%= 1.82(1.110-2.983)) - moderate malnutrition (AOR and 95%= 2.594 (1.251-5.377)) - mild malnutrition (AOR and 95% =14.25 (7.074-28.742)) - WHO stage I and II (AOR and 95% CI =3.345 (1.652-6.772)) - no opportunistic infection during admission (AOR and 95% =2.791(1.047-7.439))
Mallewa, et al 2018 31, Africa	Randomized clinical trial	-Adults -At least five years with HIV infection and CD4 cell count fewer than -100	1805 -897 to treatment -908 to non RUSF	Peri-urban regional hospital **Food type**: RUTF and RUSF	Adults and adolescents aged 13 years and older in the RUSF group had significantly greater gains in BMI, and MUAC than the no-RUSF group	Not assessed

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		-Starting ART at the health facilities			(p=0·004, and 0·03, respectively). **Measurement:** Improvement in weight, BMI and MUAC (no cut off point)	
Nagata et al 2014 40, Kenya	Retrospective cohort study	-non pregnant adult patients -BMI less than 18.5 kg/m^2	1017	Rural village **Food type**: fortified corn-soy flour supplements	On average, enrolled clients gained 2.01 kg in weight and 0.73 kg/m^2 in BMI over follow-up (mean 100 days) with the highest gain among those severely malnourished. - Only 13.1% nutritional recovery according to criteria **Measurement**: BMI.20 kg/m^2	The factors associated with BMI >20 Kg/m^2 were - Greater BMI at baseline 1.52 (1.23–1.86) - younger age, 0.97 (0.94–1.00) - male gender, (0.51 (0.30–0.86)) - not requiring highly active antiretroviral therapy (HAART) (4.18 (2.42–7.22)

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Ndekha et al 2009 34, Malawi	Randomised control trial	- Adults living with HIV - BMI<18.5 Kg/m^2	491	Clinic based **Food type**: fortified corn-soy flour supplements	After 14 weeks, patients receiving fortified spread had a greater increase in BMI and fat-free body mass than those receiving corn-soy blend: 2.2 (SD 1.9) v 1.7 (SD 1.6). Measurement: Gain in BMI (no cut off point)	No significant differences in the CD4 count, HIV viral load, assessment of quality of life, or adherence to antiretroviral therapy were noted between the two groups
Oguntibeju et al 2007 41, South Africa	Retrospective cohort study	- non pregnant adults with HIV, non-diabetic and non-hypertensive - not on ART - CD4 cell count 100-350 cells/mm^3	35	Not specified **Food type**: extract of hypoxis, grape fruit seed extract, sitosterol and sitosterolin, beta-carotene, vitamin E, vitamin, vitamin B1, vitamin B2, vitamin B12, nicotinamide, vitamin C, olive green	19 (67.9%) of the 28 patients had a BMI within the normal range after nutrient intervention **Measurement:** BMI>=18.5kg/m^2	- significant positive correlation between the BMI and fat percentage (statistics not reported) - CD4+T cell count showed no correlation with any of the anthropometric indices - high viral load has significant negative correlation with the lean body mass and BMI (statistics not reported)

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				leaf extract, folic acid and natural antioxidant (biocydin)		
Olsen et al 2014 33, Ethiopia	Randomised control trial	- Age greater than 18 years - BMI greater than 16 Kg/m^2	453	Referral hospital **Food type**: Plump sup	At three months, participants receiving the supplements containing whey or soy had increased their lean body mass by 0.85 kg (95% confidence interval 0.16 kg to 1.53 kg) and 0.97 kg (0.29 kg to 1.64 kg), respectively, more than controls. **Measurement**: Weight gain and BMI (no cut off point)	- No effect on physical activity - The effects of the two supplements, however, were not significantly different in direct comparison
Rawat et al, 2014 35, Uganda	Prospective cohort	- Food insecure people living with HIV who are new to ART. - employment status and income	904	Community based **Food type**: WFP Rations	Over 12 months, food assistance significantly increased body mass index by 0.6 kg/m2 (P, 0.01) and mid-upper arm	- no impact on CD4 count, Hb concentrations, or Individual Dietary Diversity Score - at the household level, food assistance increased the Household Food Insecurity Access Scale, by 2.1 points (P, 0.01) - food assistance increased BMI by 0.6 kg/m2 and MUAC by 6.7 mm.

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		- housing characteristics and past experience of food insecurity			circumference by 6.7 mm (P, 0.05). **Measurement:** Change in BMI (no cut off points)	- no statistically significant impact of food assistance on Hb concentration or CD4 count
Sadler, et al, 25 Ethiopia	Comparative follow up study. Report, not peer reviewed	Adults living with HIV - BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 intervention were included in the final analysis	Health facilities / health centres **Food type**: RUTF/RUSF	636 (32.6) graduated in the intervention group and 120 (18.8) in the comparison group **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Factors that were associated with nutritional recovery were: - being in the intervention group 2.4 (1.6-3.4) - female 1.4 (1.0-1.9) - attending in a hospital 0.5 (0.4-0.7)
Scarcella. et al 43, Mozambique	Randomised control trial	- Adult HIV patients - BMI <18.5 Kg/m^2 - presence of wasting - TB co-infection - insufficient access to food - CD4 count < 200	106	Public Health Centre **Food type**: - food package consisted of a ration of cereals (rice or corn) of around 15kg; pulses (5kg); peanuts (3kg); sugar (2kg); sunflower seed oil (3Lt.);	In the cohort not in program the mean BMI values showed an increase but not significant (initial value: 21.9±2.9; final value: 22.5±3.3) **Measurement**: Increase in BMI (no cut off point)	- haemoglobin showed an increase but not significant - in the intervention group both, BMI and Haemoglobin shown a significant increase - increase in BMI was statistically associated with the dietary diversity score in the intervention group

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				at least 7kg of fortified corn soy blend		
Tesfay et al, 42 Ethiopia	Retrospective cohort study	- Adults living with HIV - BMI<18.5	1757	Hospitals **Food type**: RUTF/RUSF	55.3% (95% CI = 53.2–57.4) recovered from malnutrition **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Not assessed
van Oosterhout et al 2010 37, Malawi	Retrospective cohort study	-Adults aged 18 years and older -BMI 18.5 kg/m^2 -ART naïve or - WHO clinical stage III or IV Or - CD4 count B250 cells/ml irrespective of clinical stage	593	Centeral hospital **Food type**: corn/soy blended flour (CSB) or ready-to-use fortified spread (RUFS), a peanut butter-based product	- Linear regression analysis showed that increase in BMI was greatest in the RUSF group and better in the CSB group than in those receiving no food supplementation at 14 weeks. - These differences were no longer significant at 26 weeks **Measurement:** Increase in BMI	- no longer a significant effect of supplementary feeding on an increase in BMI, 12 weeks after the feeding was stopped. - factors that were associated with nutritional recovery or improved BMI was not assessed.

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Annex table 2: Shows program failure (default + nonresponse +death) after enrolment in a nutritional program in sub Saharan Africa

Author, year and country	Study design	Study population	N	Setting and type of food	Nutritional program failure	Factors that affect nutritional program failure
Ahou et al 2011 38, Kenya and uganda	Retrospective cohort study	Adults aged 15 years and older living with HIV BMI less than 17 kg/m^2 and bilateral pitting oedema	1106	Regional hospital Type of food: RUTF/RUSF	An overall total of 531 of (48.0%) patients failed nutrition therapy	Factors that were associated with program failure were: - male gender (OR = 1.5, 95% CI 1.2-2.0), - patients with severe malnutrition at nutrition programme enrolment (OR = 2.2, 95% CI 1.7-2.8), and - those never started on ART (OR = 4.5, 95% CI 2.7-7.7) for those eligible; OR = 1.6, 95% CI 1.0-2.5 for those ineligible for ART at enrolment
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital Type of food: RUTF/RUSF	Overall program failure rate was 196 (37.6%)	The factors that were associated with nutritional program failure were not assessed.

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Annex table 3: Nonresponse to the nutritional program and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food	Nonresponse to the nutritional program	Factors that contribute to non-response to the nutritional program
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUFT/RUSF	- 155 (29.6%) non-respondent - **Measurement**: not reaching 18.5 kg/m^2	The factors that affect non-response were not assessed
Sadler, et al , 25 Ethiopia	Comparative follow up study. not peer reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each. 1956 and 639 were including in the final analysis	Health facilities / health centres **Food type**: RUFT/RUSF	- 1316 (67.4%) non-respondent in the intervention group and 519 (81.2) become non-respondent in the comparison group - **Measurement**: not reaching 18.5 kg/m^2	Factors that were associated with nutritional non-response were: - attending the program at a hospital was associated with an increased likelihood of non-response (OR=1.75, p=0.021), - severe food insecurity was associated with a decreased likelihood of non-response (OR=0.46, p=0.016) compared to those who were "food secure.
Tesfay et al , 42 Ethiopia	Retrospective cohort study	Adults living with HIV and-BMI<18.5	1757	Hospitals **Food type**: RUFT/RUSF	- 21% (95% CI = 19.7–23.4) completed the program but failed to recover from malnutrition. - **Measurement**: not reaching 18.5 kg/m^2	- urban residence, (AOR = 1.46, 95% CI = 1.4–2.91), - attending Shul (AOR = 2.92, 95% CI = 2.04–4.19) and - Lemlem Karl (AOR = 1.49, 95% CI 1.05–2.11) hospitals

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Annex table 4: Default from the nutritional program and its determinants in sub-Saharan Africa

Author, year, and country	Study design	Study population	N	Setting	Default to the nutritional program	Factors that contribute to default to the nutritional program
Sadler, 25 et al , Ethiopia	Comparative follows up study. not peer-reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 intervention were included in the final analysis	Health facilities / health centres	70.6% defaulted from the nutritional program among the intervention group	- severe acute malnutrition (OR: 1.89, P-value=0.003)
Tesfay et al 42 , Ethiopia	Retrospective cohort study	Adults living with HIV and BMI<18.5	1757	Hospitals	19% (95% CI, 17.3–20.7) did not complete the program.	Factors associated with program incompleteness. - living in urban areas (AOR = 1.44, 95% CI = 1.05– 1.97), - employed (AOR = 1.39, 95% CI = 1.01–1.93), - attending Shul (AOR = 4.6, 95% CI = 3.15–6.71) and Lemlem Karl (AOR = 2.5, 95% CI = 1.69–3.71) hospitals, - in clinical stages II (AOR = 2.49, 95% CI = 1.59–3.91) and III (AOR = 1.46(1.02–2.07), - on ART for less than six months (AOR = 1.61, 95% CI = 1.09–2.39), - anaemic (AOR = 1.77, 95% = 1.29–2.41), and - severe acute undernutrition at enrolment (AOR = 6.43, 95% CI = 4.69–8.3)

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Supplementary file 1: Shows the searching terms and outputs for Medline

#	Searches	Results
1	HIV Infections/ or HIV/ or HIV-2/ or HIV-1/	229102
2	Acquired Immunodeficiency Syndrome/	75776
3	(HIV* or AIDS* or "Acquired Immunodeficiency Syndrome").tw,kf.	382368
4	1 or 2 or 3	420727
5	Nutritional Status/	41768
6	Dietary Supplements/	52953
7	Food Assistance/ or *Food/ or *"Diet, Food, and Nutrition"/ or Food, Fortified/	28061
8	nutrition therapy/ or diet therapy/ or nutritional support/	18224
9	((food or nutrition) adj3 (support or program or intervention or supplement* or therap* or care or consult*)).tw,kf.	24448
10	5 or 6 or 7 or 8 or 9	151790
11	"africa south of the sahara"/ or africa, central/ or cameroon/ or central african republic/ or chad/ or congo/ or "democratic republic of the congo"/ or equatorial guinea/ or gabon/ or "sao tome and principe"/ or africa, eastern/ or burundi/ or djibouti/ or eritrea/ or ethiopia/ or kenya/ or rwanda/ or somalia/ or south sudan/ or sudan/ or tanzania/ or uganda/ or africa, southern/ or angola/ or botswana/ or lesotho/ or malawi/ or mozambique/ or namibia/ or south africa/ or swaziland/ or zambia/ or zimbabwe/ or africa, western/ or benin/ or burkina faso/ or cabo verde/ or cote d'ivoire/ or gambia/ or ghana/ or guinea/ or guinea-bissau/ or liberia/ or mali/ or mauritania/ or niger/ or nigeria/ or senegal/ or sierra leone/ or togo/	197050
12	("africa south of the sahara" or africa*or cameroon or central african republic or chad or congo or "democratic republic of the congo" or equatorial guinea or gabon or "sao tome and principe" or burundi or djibouti or eritrea or ethiopia or kenya or rwanda or somalia or south sudan or sudan or tanzania or uganda or angola or botswana or lesotho or malawi or mozambique or namibia or "south aftrica" or swaziland or zambia or zimbabwe or benin or burkina faso or "cabo verde or cote d'ivoire" or gambia or ghana or guinea or guinea-bissau or liberia or mali or mauritania or niger or nigeria or senegal or sierra leone or togo).mp.	361692
13	11 or 12	412824
14	4 and 10 and 13	616
15	limit 14 to english language	604
16	limit 15 to yr="2005 -Current"	510

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Supplementary file 2: JBI Critical Appraisal Checklist for Quasi-Experimental Studies (non-randomized experimental studies)

Reviewer_____Date_____

Author_____Year_____Record Number _____

	Yes	No	Unclear	Not applicable
1. Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	☐	☐	☐	☐
2. Were the participants included in any comparisons similar?	☐	☐	☐	☐
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	☐	☐	☐	☐
4. Was there a control group?	☐	☐	☐	☐
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	☐	☐	☐	☐
6. Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	☐	☐	☐	☐
7. Were the outcomes of participants included in any comparisons measured in the same way?	☐	☐	☐	☐
8. Were outcomes measured in a reliable way?	☐	☐	☐	☐
9. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

_____

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_____

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Supplementary file 3: Assessment of methodological quality (n=16).

Authors	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	%
Ahoua, et al. (2011)	Y	N/A	N/A	Y	Y	Y	Y	Y	Y	100
Bahwere, et al. (2009).	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Berhe, et al. (2014)	Y	N/A	N/A	N	Y	Y	Y	Y	Y	86
Diouf, et al. (2016).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Mallewa, et al. (2018).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Manary, et al. (2010)	Y	Y	Y	Y	Y	Y	Y	Y	UC	100
Nagata, et al. (2014)	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Ndekha, et al. (2009)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Oguntibeju,et al. (2007)	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Olsen, et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Rawat, R., et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Rawat, et al. (2010).	Y	N/A	N/A	N	Y	Y	Y	Y	Y	86
Sadler, et al. (2012).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Scarcella, et al. (2011)	Y	Y	Y	N	Y	Y	Y	Y	Y	89
Tesfay, et al. (2020).	Y	Y	Y	N	Y	Y	Y	Y	Y	89
van Oosterhout, et al.	Y	Y	Y	Y	Y	Y	Y	Y	Y	100

Q= Question Y=Yes; N= No; NA= Not applicable, UC=unclear

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	N/A

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	N/A
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	26
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	30
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	30
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	No funding

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Characteristics and impacts of nutritional programs to address undernutrition of adults living with HIV in sub-Saharan Africa: A systematic review of evidence.

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-047205.R3
Article Type:	Original research
Date Submitted by the Author:	10-Dec-2021
Complete List of Authors:	Tesfay, Fisaha; Mekelle University College of Health Sciences, School of Public Health Javanparast, Sara; Flinders University, College of Medicine and Public Health Gesesew, Hailay ; Flinders University, College of Medicine and Public Health Mwanri, Lillian; Flinders University, College of Medicine and Public Health Ziersch, Anna; Flinders University, Southgate Institute for Health, Society and Equity
Primary Subject Heading:	HIV/AIDS
Secondary Subject Heading:	Epidemiology, Health services research, HIV/AIDS
Keywords:	PUBLIC HEALTH, NUTRITION & DIETETICS, EPIDEMIOLOGY, Infectious disease/HIV < NEUROLOGY

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Characteristics and impacts of nutritional programs to address undernutrition of adults living with HIV in sub-Saharan Africa: A systematic review of evidence.

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Abstract

Objectives: Although some studies have identified various challenges affecting nutritional programs to effectively tackle undernutrition amongst people living with HIV, evidence about the characteristics and impacts of these programs on weight-related nutritional outcomes varies based on country contexts, specific program goals and the implementation processes. This systematic review sought to synthesise evidence on the characteristics and impact of nutritional programs on weight-related nutritional outcomes of people living with HIV in sub-Saharan Africa.

Design: Systematic review

Data sources: We searched for primary studies published in the following databases: Web of Science, Medline, Scopus, ScienceDirect, ProQuest and Google Scholar, supplemented by checking reference lists of included papers.

Eligibility criteria: Studies published from 2005 to the 10th of July 2020 and reporting on the weight-related nutritional outcomes of undernourished people enrolled in nutritional programs in HIV care in Sub-Saharan Africa were included.

Data extraction and synthesis: Data were extracted using a data extraction proforma. Weight-related nutritional outcomes of people living with HIV before and after enrolment in a nutritional program were compared and narratively synthesised.

Results: Sixteen studies assessing the impact of nutritional programs in HIV care on weight-related nutritional outcomes were included**.** Of these, 13 examined nutritional programs implemented in health facilities and the remaining three were delivered outside of health facilities. Nutritional recovery (defined differently in the studies) ranged from 13.1% to 67.9%. Overall program failure rate, which included default after enrolment in a nutritional program or non-response, ranged from 37.6% to 48.0%. More specifically, non-response to a nutritional program ranged from 21.0% to 67.4% and default from the program ranged from 19.0% to 70.6%. Key sociodemographic, clinical, and nutritional characteristics that affect nutritional recovery, non-response and default were also identified.

Conclusions and recommendations: Nutritional programs in HIV care have led to some improvements in weight-related nutritional outcomes amongst people living with HIV. However, the programs were characterised by a high magnitude of default and non-response.

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To improve desired weight-related nutritional outcomes of people living with HIV, a holistic approach that addresses longer-term determinants of undernutrition is needed.

Keywords: Nutritional program, nutritional support, HIV and nutrition

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Strengths and limitations

- This review is the first of its kind to synthesise the evidence with regards to weight-related outcomes of nutritional programs to address undernutrition for people with HIV across sub-Saharan Africa, which has the largest number of people living with HIV.
- The review assesses multiple weight-related nutritional outcomes and presents a summary of evidence from a range of nutritional programs.
- However, the diversity and heterogeneity of the studies and nutritional programs meant that meta-analysis was not possible to be done, a shortcoming of the review.
- This study only focused on papers published in English which may have excluded potential studies on the same topic but published in other languages.

The original protocol for the study: The study protocol is attached at the end of the study.

Funding statement: Not funded.

A competing interest statement. All authors declare that they have no conflict of interest.

Introduction

Evidence demonstrates the clinical and social susceptibility of people living with HIV to undernutrition and its wider effects [1]. Inadequate food intake due to lack of appetite, malabsorption, and impaired storage and metabolism are clinical causes of undernutrition among people living with HIV [2], while social factors such as poverty, gender inequality, food insecurity, stigma, and discrimination also hinder access to adequate food and nutrition [3]. Undernutrition involves weight-related stunting (low height for age), wasting (low weight for height), underweight (low weight for age) and micronutrient deficiencies but this particular review will focus on undernutrition [4]. Weight related undernutrition (hereafter undernutrition) has various impacts on the wellbeing of people living with HIV, including on HIV-related mortality in resource poor settings which has been highlighted in a range of studies [5]. Research has also indicated that undernutrition can act as a barrier to the utilisation of HIV care services and antiretroviral therapy (ART) adherence [6].

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The World Health Organisation (WHO) highly recommends the integration of nutritional interventions into HIV care and support services [7]. To address the impact of undernutrition among people living with HIV, nutritional programs targeting people living with HIV are operational in resource-poor settings, mainly in sub-Saharan Africa [8]. The key elements of these programs are nutritional assessment, counselling, and nutritional support [9,10]. The latter component includes the provision of micronutrient, macronutrient, or fortified macronutrient supplementation [11,12].

Concept 1: ***Nutritional program***	**Concept 2:** ***HIV/AIDS***	**Concept 3:** ***Sub-Saharan Africa***
Nutritional program	HIV-1	South of Saharan Africa

Nearly 90% of HIV care sites in sub-Saharan Africa have responded to the WHO recommendations and instituted some forms of nutritional program [13], with substantial variations between and within countries [10,11,14]. Donor organisation policies regarding the nutritional program and availability of national and local resources have led to program differences observed between countries in sub-Saharan Africa [15], including the type of care [16,17] and nature of nutritional support provided [18]. Evidence regarding the impact of nutritional programs on nutritional outcomes is fragmented, mixed, and varies by country context, program goals, and implementation. This study aims to systematically review and synthesise evidence on the characteristics and impact of nutritional programs on nutritional outcomes of undernourished people enrolled in nutritional programs in HIV care in sub-Saharan Africa.

Methods

Protocol

The protocol for this systematic review has been registered under the PROSPERO review registration number: CRD42020196827.

Systematic search strategy

This review was guided by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA). Studies published in English language in the Web of Science, Medline, Scopus, ScienceDirect, ProQuest, from 2005 to the 10th of July 2020 were searched. Google Scholar and the reference lists of references of selected papers were also searched. The search concepts and terms included “Nutritional programs”, “HIV/AIDS”, and “sub-Saharan Africa” (**Table 1**). Outputs of Ovid Medline are shown in the **supplementary file 1**. **Table 1:** Concepts and their respective keywords

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Nutritional intervention	HIV-2	Africa
Nutritional care	HIV infection	Western Africa
Food by prescription	Human immunodeficiency syndrome	Central Africa
Supplementary food	Acquired immune deficiency syndrome	
Ready to use Therapeutic food		
Food supplementation	AIDS	List of all countries in sub-Saharan Africa

Selection of studies

The screening process involved two stages. The first stage was conducted by FT and involved screening the studies by title and abstracts against the inclusion criteria. The second stage was conducted by two independent reviewers (FT and HAG) and involved full-text screening of articles. The inclusion criteria were:

- longitudinal studies (cohort and interventional) that examined nutritional outcomes of undernourished people (age>15 years) who were enrolled in nutritional programs in HIV care settings.
- studies measuring nutritional outcomes using Body Mass Index, (BMI), weight gain or Mid Upper Arm Circumference (MUAC)
- studies published since 2005, because most nutritional programs in sub-Saharan Africa started in 2005 and published in English.

Studies involving children and pregnant women were excluded because different enrolment and nutritional outcome criteria are used to enrol children and pregnant women in programs. Studies were also excluded from the review if information about age or pregnancy status was not provided (**Figure 1**).

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Measurement of exposures

The exposure variables included:

- sociodemographic factors: age, sex, residence, marital status, and educational status.
- clinical factors: WHO clinical stage, CD4 count, functional status, presence of opportunistic infections, and HIV status disclosure.
- nutritional factors: baseline nutritional status, the status of food security and other dietary information.

Measurement of outcomes

The outcome variables were nutritional outcomes of undernourished adults living with HIV, compared before and after enrolment in a nutritional program. Outcomes included changes in BMI, weight, or MUAC and nutritional recovery (BMI$\geq$ 18.5 kg/m^2), default, or non-response to the nutritional program.

Context/setting

Most nutritional programs in sub-Saharan Africa commenced in 2005. Most nutritional programs in sub-Saharan Africa included in the reviewed studies had similar program components which included nutritional assessment, counselling, and nutritional support. This review is limited to the sub-Saharan Africa given the duration, enrolment, and exit criteria of nutritional programs are very different in other regions and settings.

Assessment of methodological quality

The Joanna Briggs Institute [19] for critical appraisal tool for quasi-experimental studies was used to assess the quality of studies that were included in the review. Two independent reviewers (FT and HAG) conducted the quality assessment.

Data extraction

Data were extracted using a proforma adopted from JBI-MAStARI. The extracted data involved information about the outcome and exposure that affected the nutritional outcomes. In addition, population, sample size, country description of the nutritional program, and the conclusion of the studies were extracted.

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Data analysis

Due to the heterogeneity of the studies included in terms of study design, the measure of nutritional outcomes used, and differences in enrolment and exit criteria, a narrative synthesis was the most appropriate analysis procedure for this review. Different variables were identified from the reported findings of each study. Similar patterns were grouped in subsequent coding and analysis performed. Finally, the findings were categorised into two major themes.

Patient and Public Involvement: No patients or public was involved in this study. This is a systematic review and it is not applicable to make patient and public involvement.

Results

Description of the studies

A total of 2036 papers were obtained from electronic databases and other sources. After removing duplicates 1,738 studies were screened by title and abstract and 1,708 were deemed irrelevant. The remaining 30 papers were scrutinised further against the inclusion criteria using full-text screening. Sixteen papers involving 25,335 participants were included in the final review (**Figure 1**). Of the 16 final studies included in the review, nine were retrospective cohort studies, three were prospective cohort studies, and four were randomised trials.

Assessment of methodological quality

The methodological and reporting quality of the reviewed studies was assessed using the JBI critical appraisal tool for quasi-experimental studies (**supplementary file 2**). The assessment tool included nine parameters each carrying equivalent weight (**supplementary file 3**).

Nine studies, six clinical trials [20-25], and three cohort studies [17,26,27] scored nine out of nine (100%).

One retrospective cohort study [28] scored seven out of nine but two parameters (related to the comparability of study participants and similarity of interventions in both the exposed and non-exposed groups) were not applicable− scoring 100%. Three cohort studies scored six out of nine which is 100% as the same two parameters were not applicable, and the presence of control groups was not clear [29-31].

Two retrospective studies scored eight out of nine (89%) as there was no control group [32,33]. Another retrospective cohort study scored six out of nine, with no control group and two parameters not applicable - a final score of 86% [34].

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Characteristics of the nutritional programs

The nutritional programs examined in this review varied in terms of enrolment criteria, duration, and type of nutritional support used. Six studies used BMI <18.5 kg/m^2 as an enrolment criteria [17,23,24,30,32,34]. Another seven studies used one of the following: bilateral oedema, being bedridden for 50% of the month, TB coinfection, CD4 count < 350, WHO clinical stage III and IV, in addition to BMI< 18.5 kg/m^2 or MUAC <210mm) [21,22,27-29,31,33]. Living in a household with HIV positive family members was used as an enrolment criterion in one study [26]. Another study enrolled people living with HIV at any stage of the disease and not able to eat [20]. A further study used the world food program (WFP) poverty assessment index, not receipt of other food assistance, CD4 count between 200 and 450 cells per microliter, and ART naïve (new for the treatment) as an enrolment criterion [25].

In five studies, the duration of the nutritional program ranged from three months (for mild and moderate undernutrition) to six months (for severe undernutrition) [17,28,29,32,34]. Two studies reported nine weeks of program duration 33, 44, two had 14 weeks [22,24], and a further study [21] 24 weeks duration before individual patients exited from the program. The nutritional program was nine months in one study [27] and twelve months in three studies [23,25,26]. No duration limit was reported in two studies [30,33].

Various types of nutritional supports were used in nutritional programs. Ready to use therapeutic food (RUTF) or ready to use supplementary food (RUSF) were used in eight studies [17,20,21,23,28,29,32-34]. Corn/soy blended food was used in three studies [22,24,27,30] and the WFP food basket (which involves generic food rations) was used in two studies [25,26]. One study used an extract of hypoxis, grapefruit seed extract, sitosterol and sitosterolin, beta-carotene, vitamin E, vitamin, vitamin B1, vitamin B2, vitamin B12, nicotinamide, vitamin C, olive green leaf extract, folic acid, and natural antioxidant (biocydin) as the nutritional supplement [31].

Although nutritional assessment was an element of nutritional programs in all studies included in this review, the provision of nutritional counselling was stated only in two studies [23], [32]

The nutritional programs were provided in health facilities such as hospitals and clinics in 12 studies [17,20-24,26-28,32-34], in community-based settings in four studies [23,29,30], and no setting was specified in one study [31].

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Impact of the nutritional program on weight related nutritional outcomes (nutritional outcomes here after)

In reporting improvements in the nutritional outcome of people living with HIV, studies used various outcome measures. Gain in BMI, weight, and MUAC were used either in combination or as separate nutritional outcome measures. BMI was used in nine studies [17,24,25,27,28,31-34] and both MUAC and BMI were used in two studies [21,29]. The other two studies used weight gain [20,26], and three studies used both BMI and weight gain to measure the nutritional outcomes [20,23,26].

Nutritional program failure, recovery, default, and non-response were main outcomes measured in most review studies. The definition of these terms in the studies included are shown in **table 2.**

Table 2: The definition of nutritional outcomes in nutritional programs

Outcomes	**Adults**
Program failure	Includes default and non-response from the nutritional program
Improvement in nutritional status and recovery /graduation from the nutritional program	Various criteria were used to determine nutritional recovery • Reaching BMI of 18.5kg/m^2 for two consecutive visits in some studies or BMI>18 kg/m^2 or BMI>20kg/m^2 In other studies, the nutritional outcome was measured in terms of improving weight, BMI, or MUAC.
Non-response	Participants did not reach the target BMI for two consecutive visits or did not show any change in BMI, weight, and MUAC
Default /failure to complete the nutritional program	Dropped out of the program before reaching the target BMI, weight or MUAC

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Improvement in nutritional status or recovery from the nutritional program

Eleven studies measured nutritional recovery which was defined differently in various nutritional programs. Three studies defined program recovery as BMI>18.5kg/m^2 in two consecutive measurements [17,32,34], and nutritional recovery in these studies ranged from 32.6% [17] to 62.2% [34]. One study used BMI >18 kg/m^2 in two subsequent measurements and the absence of oedema, and recovery was 47.4% [28]. A further study defined program recovery as BMI>20kg/m^2 and 13.1% attained nutritional recovery against these criteria [30].

Four studies used improvements in BMI, without specifying a specific cut-off point [24,25,27,33]. The improvement in BMI in these studies ranged from 0.6 kg/m^2 using WFP rations for 12 months [25] to 2.2 kg/m^2 in a study where participants received a fortified spread for 14 weeks [24] and those who used a corn-soy blend showed a BMI increase of 1.7 kg/m^2 [24]. Weight gain was used to measure nutritional outcomes in four studies [20,23,26,29] and weight gain ranged from 0.85 kg [23] to 3 kg [29].

One study using weight gain found that after 9 weeks of supplementation, body weight, and fat-free mass increased significantly by 11 % (p= 0.033), and 11.8 % (p= 0.033) in the RUTF group, but not in the control group [20]. One study used BMI, weight gain, and MUAC [21], and those in the RUSF group had significantly greater gains in weight, BMI, and MUAC than those in the no-RUSF group (p=0·004, and 0·03, respectively).

Annex table 1 presents the socioeconomic, demographic, clinical, and nutritional determinants of nutritional outcomes among those enrolled in nutritional program. For sociodemographic factors that affected nutritional outcomes, female gender was associated with nutritional recovery in three studies [17,30,34]. Compared with those who didn't attend any schooling, individuals who attended school were also more likely to recover than those who did not in one study [34], and attending the nutritional program in a hospital was inversely related to nutritional recovery in another study [17].

Few clinical characteristics were associated with nutritional recovery. In one study, not being on ART was associated with nutritional recovery [30]. In another study, compared with WHO clinical stage 4, being at WHO clinical stage 1 and 2 were associated with nutritional recovery [34]. In addition, the absence of opportunistic infection at enrolment to the nutritional program was associated with nutritional recovery compared with those who had an opportunistic infection [34].

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Regarding the impact of nutritional characteristics at enrolment, compared with those enrolled with severe undernutrition, those with moderate undernutrition were associated with nutritional recovery in two studies [30,34]. In one study, enrolment in the nutritional program with mild undernutrition was associated with nutritional recovery compared with those with severe undernutrition [34].

Program failure rate

The program failure rate involves a combination of default, non-response, and mortality while enrolled in the nutritional program. The program failure rate was reported in two studies [28,34]. The nutritional program failure rate in a study from Kenya and Uganda was 48% [28] and 37.6% in a study from Ethiopia [34]. Factors that were associated with the program failure were only assessed in one study [28] which found male gender, severe undernutrition at program enrolment (pre-ART), and ineligibility for ART at enrolment as variables associated with program failure (**Annex table 2**).

Non-response from the nutritional program

Three studies from Ethiopia measured non-response to the nutritional program ranging from 21% [32], to 29.6% [34], and 67.4% [17].

Compared with those who attended primary health care services to receive nutritional services, attending a hospital was associated with greater non-response in one study [17]. A study by Tesfay and colleagues examining health facility based nutritional programs in Ethiopia reported that being an urban area was associated with non-response [32].

In terms of clinical characteristics, the same study found advanced WHO clinical stage (stage IV), and bedridden functional status were associated with non-response to the nutritional program [32]. Compared with those enrolled with mild undernutrition, severe undernutrition at enrolment was also associated with non-response [32].

In another study, those with severe food insecurity were less likely to not respond than those who were food secure [17]. The third study did not assess factors contributing to non-response (**Annex table 3**).

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Default from the nutritional program

Two studies from Ethiopia reported program default and associated factors.

In one study, 19% of the study participants did not complete the nutritional program. Factors that were associated with default were living in urban areas, being employed, attending at a hospital, and in clinical stages II and III [32]. In another study, the default rate was 70.6% [17]. Being on ART for less than six months and being anaemic were associated with default [17] (**Annex table 4**).

In both studies, severe undernutrition at enrolment to the nutritional program was associated with default from the nutritional program [17,32].

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Discussion

Given the susceptibility of people living with HIV to undernutrition, and the synergistic effect of undernutrition in accelerating HIV infection to AIDS, scaling up nutritional programs in HIV care in combination with other HIV services is vital [35]. Despite the provision of nutritional programs in HIV care settings in many countries [14], evidence regarding the effectiveness of these programs is fragmented–this review sought to address this with regards to sub-Saharan Africa.

Broadly, a range of anthropometric, clinical, and socioeconomic parameters were used as enrolment criteria in nutritional programs in the HIV care setting in sub-Saharan Africa. The duration of the nutritional programs was characterised by a significant variation between programs. For instance, nutritional programs that aimed at the treatment of acute undernutrition were characterised by a shorter duration (ranging from three to six months) and provision of highly dense nutritional support (RUTF or RUSF) [17,28,32]. However, there were examples where nutritional support was provided for more than nine months [21,22,24]. The duration of nutritional programs that used corn-soy blended ranged from 14 weeks [22,24] to 9 months [27], and 12 weeks in programs that used the WFP food rations [25].

There was no optimal duration of the nutritional program because of the difference in content and nutritional value of the nutritional support used from program to program. Nutritional programs that use dense nutritional supports such as RUTF and RUSF may achieve faster weight gain and perhaps contribute to a shorter duration in the nutritional program [17,32]. These findings are consistent with outcomes from other studies conducted in other parts of the world such as India [36].

There is strong evidence in the literature regarding the role of nutritional programs in enhancing weight gain among those enrolled in a nutritional program with BMI<18.5 kg/m^2 [37]. However, in nutritional programs with predetermined program exit criteria, there was substantial variation in the magnitude of nutritional recovery. Nutritional recovery beyond attaining weight gain can vary from program to program depending on the availability of resources. In nutritional programs that used BMI>18.5kg/m^2 in Ethiopia, nutritional recovery was 32.6% [17], 55.3% [32] and 62.2% [34]. Other studies did not use a defined cut-off point or used different BMI measure (e.g., one used BMI>18 and another used BMI>20) or other measures (weight gain, MUAC) to determine nutritional outcomes. All of these demonstrated that nutritional support of various forms were associated with better nutritional outcomes.

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A systematic review that examined the impact of nutritional supplementation on HIV nutritional and clinical prognosis in sub-Saharan Africa found increased weight gain [38].

Evidence from the current review showed that few studies reported on the rate of overall nutritional program failure. Two studies from Ethiopia found program failure rates involving non-response, default, and mortality rates that ranged from 37.6% [34] to 48.0%[28].

Studies examining non-response to the nutritional program (not reaching a BMI of 18.5kg/m^2 in two consecutive visits) were all from Ethiopia. While the nutritional program was similar in terms of the duration, type of food supplement, funding organisation, and policy, there was a big variation in the magnitude of non-response rate between these studies of 21% [32], 29.6% [34], and 67.4% [17]. The two studies [32,34] with a relatively low non-response rate were conducted recently, after several years of implementation of the nutritional program in Ethiopia, which may suggest that program non-response rates may have dropped due to the accumulation of experience by the program.

Two studies from Ethiopia measured default from the nutritional program and the difference in default between the two studies was substantial −ranging from 19% [32] to 70.6 % [17]. While the study by Tesfay et al (2020) was conducted recently, the later study by Sadler (2012) was conducted at the initial stage of the nutritional program as a pilot evaluation and may reflect a less developed version of the program.Reasons for nutritional program ineffectiveness could include poor taste of the supplement, dietary boredom, or HIV-related complications such as thrush as highlighted in a systematic review that involved resource adequate and constrained settings [39].

It is very difficult to compare nutritional outcomes by country because of the lack of uniformity in the nature of the nutritional program in terms of the duration, enrolment criteria, and the nutritional support that is given. For instance, nutritional programs in Ethiopia used ready to use therapeutic or supplementary food for three months for mild and moderately undernourished and six months for severe undernutrition [17,32,34]. On the other hand, nutritional programs from Kenya used ready to use therapeutic or supplementary foods provided in a rural village but no specific duration was given for the nutritional program [30]. Nutritional recovery (achieving BMI greater than 20 kg/m^2) in Kenya [30] was lesser (13%) than studies from Ethiopia which were 62.2% [34], and 32.6% [17].

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The studies provided some insights into factors associated with program outcomes, including socio-demographic factors. Gender was highlighted, where being male was associated with nutritional program failure [28] and lower rates of recovery [17,30,34] and living in urban areas was associated with program non-response [32]. In one study, attending the nutritional program in a hospital was associated with lower rates of nutritional recovery [17]. Another study found differences between hospitals where attending the nutritional program at relatively rural hospitals was associated with higher rates of non-response [32].

Clinical factors were also linked to program outcomes. For example, not starting ART was associated with nutritional program failure in one study [28], but another study found it was associated with nutritional program recovery [30]. This might be due to the fact that many individuals in the study by Ahoua et al 2011 were eligible for ART despite the lack of ART initiation because of delayed patient laboratory results or patient refusal. These patients might have died, giving rise to a high magnitude of program failure in the former study. Being on ART for less than six months were associated with default from the nutritional program [32]. The absence of opportunistic infection at enrolment was associated with nutritional recovery in one study [28].

Severe undernutrition at baseline was associated with a higher nutritional program failure rate [28], lower likelihood of achieving nutritional recovery [30], and default from the nutritional program [32]. On the other hand, mild and moderate undernutrition at enrolment was associated with nutritional recovery in one study [34]. Compared with those who are not enrolled in the nutritional intervention, those enrolled in the nutritional program were more likely to recover [17]. One study found that those with severe food insecurity were less likely to become non-respondents to the nutritional program [17].

Given the clinical, social, and economic susceptibility of people living with HIV to undernutrition [40], findings from the current review highlighted that nutritional programs in sub-Saharan Africa have focused on the treatment of undernutrition among people living with HIV [17,28,30]. For instance, terms such as nutritional recovery, weight gain, default, and nutritional recovery were used in all of the studies included to imply the short term positive impact of the nutritional programs among people living with HIV. Measures used in all nutritional programs were clinical and none provided measures on longer-term nutritional wellbeing. Longer-term measures such as food security were not included as key program goals in any of the nutritional programs despite the wider vulnerability of people living with HIV to food insecurity in sub-Saharan Africa [41].

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Strength and Limitations

The key strength of this study is its coverage of sub-Saharan Africa, a major area of HIV infection, and its review of multiple nutrition programs representing a large population base. It also examines a range of nutrition outcomes. However, the review has some limitations., Some studies had a very small sample size (e.g. [31] (n=35) and [33] (n=106)) which may negatively impact the estimates of the nutritional outcomes and factors associated with them. It was not possible to pool estimates of nutritional outcomes because of the differences in enrolment and exit criteria and program duration between nutritional programs. This study only included papers published in English which automatically excluded papers published in other languages, which may lead to reporting bias. The focus of this review is on the impact of nutritional programs on undernourished people enrolled in nutritional programs in HIV care settings. However, overnutrition has also been highlighted as an important issue for people living with HIV as it increases the risk of overweight and cardiovascular disease among people with HIV on long-term ART treatment [42]. An examination of programs designed to address this would be an important area for future reviews.

Conclusion and recommendations

Despite the use of various enrolment and exit criteria, nutritional programs in HIV care have led to overall improvements in nutritional outcomes and overall wellbeing of people living with HIV. However, the programs reviewed were characterised by problems of effectiveness such as a high magnitude of default and non-response. Various sociodemographic, clinical, and nutritional characteristics have an impact on the nutritional outcomes, which gives potential avenues for future research and program improvement. However, the goal of most nutritional programs in HIV care settings is to reverse short-term nutritional problems such as undernutrition. In contrast, the impact of nutritional programs in HIV care should be measured in terms of the program's capacity to sustainably address the nutritional wellbeing of individual HIV patients and others around them over the longer term [43].

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Author contributions: **FT** conceptualised, designed the study, conducted the search, analysed, and interpreted the data as well-drafted the manuscript. **HAG** assisted with the analysis. **SJ, AZ, HAG, and LM** critically reviewed the design, analysis, and interpretation of the findings and critically reviewed and approved the manuscript.

Any checklist and flow diagram: The PRISMA diagram was used.

A patient consent form: All primary studies were assessed for the fulfillment of ethical standards, and all fulfilled the ethical standards.

Data statement: The authors are happy to share the data extracted from the included papers.

Supplementary and raw data: We do not have supplementary raw data, but we are happy to share the data extracted from each included paper.

Acknowledgments: This manuscript builds on a Ph.D. thesis, and we would like to acknowledge the Australian Government Research Training program and Flinders University for supporting the study through a student support payment and fieldwork funding. We would also like to acknowledge Flinders University librarian−Shannon Brown for assisting the searching process of this systematic review.

Figure title

Figure 1: Flow diagram of literature search, selection and reason for exclusion of removed studies

Ethical Approval Statement: Not applicable. This study did not involve human subjects. Hence no ethics approval was sought

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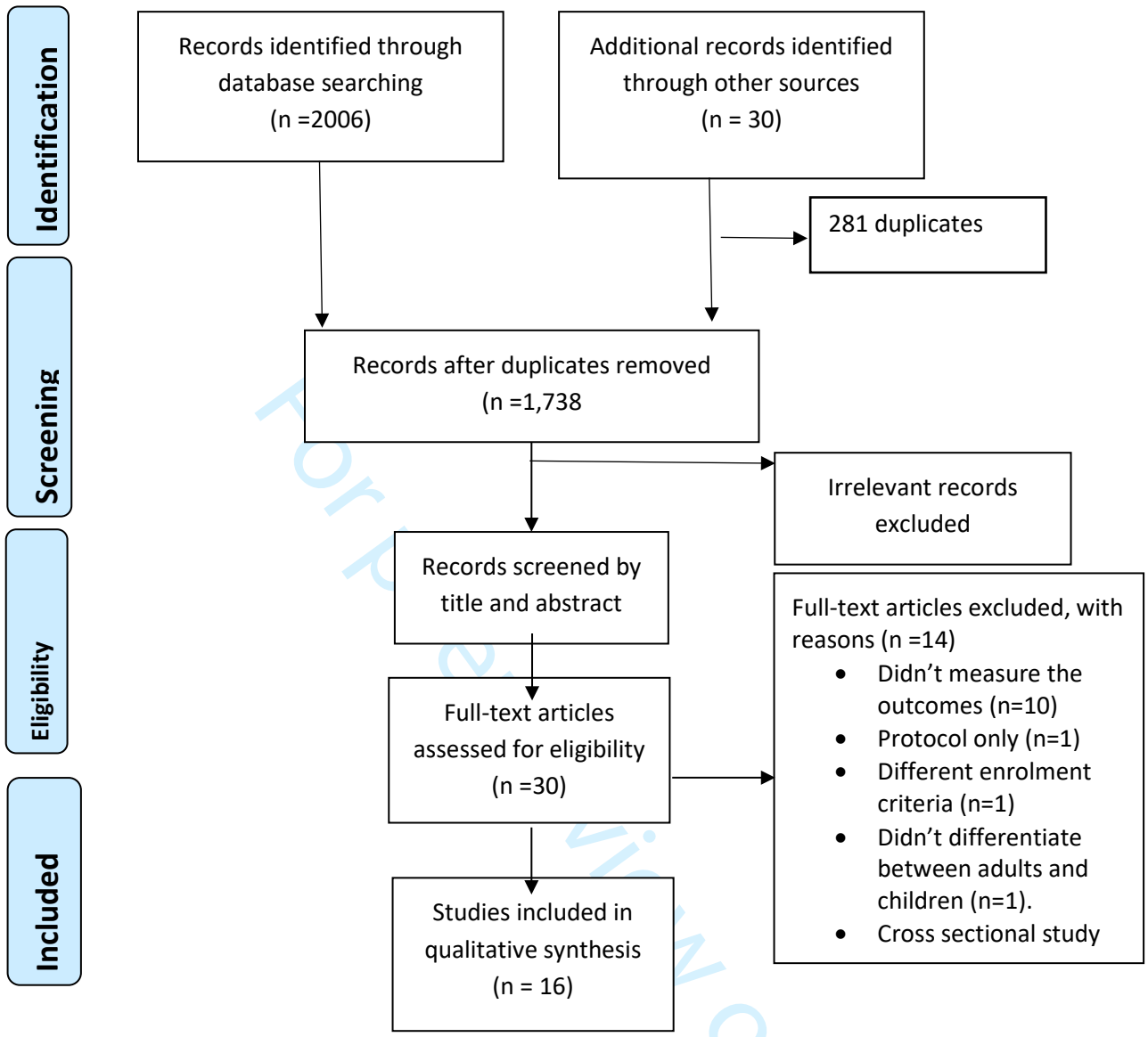


Figure 1: Flow diagram of literature search, selection and reasons for exclusion of removed studies

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Supplementary file 1: Shows the searching terms and outputs for Medline

#	Searches	Results
1	HIV Infections/ or HIV/ or HIV-2/ or HIV-1/	229102
2	Acquired Immunodeficiency Syndrome/	75776
3	(HIV* or AIDS* or "Acquired Immunodeficiency Syndrome").tw,kf.	382368
4	1 or 2 or 3	420727
5	Nutritional Status/	41768
6	Dietary Supplements/	52953
7	Food Assistance/ or *Food/ or *"Diet, Food, and Nutrition"/ or Food, Fortified/	28061
8	nutrition therapy/ or diet therapy/ or nutritional support/	18224
9	((food or nutrition) adj3 (support or program or intervention or supplement* or therap* or care or consult*)).tw,kf.	24448
10	5 or 6 or 7 or 8 or 9	151790
11	"africa south of the sahara"/ or africa, central/ or cameroon/ or central african republic/ or chad/ or congo/ or "democratic republic of the congo"/ or equatorial guinea/ or gabon/ or "sao tome and principe"/ or africa, eastern/ or burundi/ or djibouti/ or eritrea/ or ethiopia/ or kenya/ or rwanda/ or somalia/ or south sudan/ or sudan/ or tanzania/ or uganda/ or africa, southern/ or angola/ or botswana/ or lesotho/ or malawi/ or mozambique/ or namibia/ or south africa/ or swaziland/ or zambia/ or zimbabwe/ or africa, western/ or benin/ or burkina faso/ or cabo verde/ or cote d'ivoire/ or gambia/ or ghana/ or guinea/ or guinea-bissau/ or liberia/ or mali/ or mauritania/ or niger/ or nigeria/ or senegal/ or sierra leone/ or togo/	197050
12	("africa south of the sahara" or africa*or cameroon or central african republic or chad or congo or "democratic republic of the congo" or equatorial guinea or gabon or "sao tome and principe" or burundi or djibouti or eritrea or ethiopia or kenya or rwanda or somalia or south sudan or sudan or tanzania or uganda or angola or botswana or lesotho or malawi or mozambique or namibia or "south aftrica" or swaziland or zambia or zimbabwe or benin or burkina faso or "cabo verde or cote d'ivoire" or gambia or ghana or guinea or guinea-bissau or liberia or mali or mauritania or niger or nigeria or senegal or sierra leone or togo).mp.	361692
13	11 or 12	412824
14	4 and 10 and 13	616
15	limit 14 to english language	604
16	limit 15 to yr="2005 -Current"	510

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Supplementary file 2: JBI Critical Appraisal Checklist for Quasi-Experimental Studies (non-randomized experimental studies)

Reviewer_____Date_____

Author_____Year_____Record Number _____

	Yes	No	Unclear	Not applicable
1. Is it clear in the study what is the 'cause' and what is the 'effect' (i.e. there is no confusion about which variable comes first)?	☐	☐	☐	☐
2. Were the participants included in any comparisons similar?	☐	☐	☐	☐
3. Were the participants included in any comparisons receiving similar treatment/care, other than the exposure or intervention of interest?	☐	☐	☐	☐
4. Was there a control group?	☐	☐	☐	☐
5. Were there multiple measurements of the outcome both pre and post the intervention/exposure?	☐	☐	☐	☐
6. Was follow up complete and if not, were differences between groups in terms of their follow up adequately described and analyzed?	☐	☐	☐	☐
7. Were the outcomes of participants included in any comparisons measured in the same way?	☐	☐	☐	☐
8. Were outcomes measured in a reliable way?	☐	☐	☐	☐
9. Was appropriate statistical analysis used?	☐	☐	☐	☐

Overall appraisal: Include ☐ Exclude ☐ Seek further info ☐

Comments (Including reason for exclusion)

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Supplementary file 3: Assessment of methodological quality (n=16).

Authors	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	%
Ahoua, et al. (2011)	Y	N/A	N/A	Y	Y	Y	Y	Y	Y	100
Bahwere, et al. (2009).	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Berhe, et al. (2014)	Y	N/A	N/A	N	Y	Y	Y	Y	Y	86
Diouf, et al. (2016).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Mallewa, et al. (2018).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Manary, et al. (2010)	Y	Y	Y	Y	Y	Y	Y	Y	UC	100
Nagata, et al. (2014)	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Ndekha, et al. (2009)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Oguntibeju,et al. (2007)	Y	N/A	N/A	UC	Y	Y	Y	Y	Y	100
Olsen, et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Rawat, R., et al. (2014)	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Rawat, et al. (2010).	Y	N/A	N/A	N	Y	Y	Y	Y	Y	86
Sadler, et al. (2012).	Y	Y	Y	Y	Y	Y	Y	Y	Y	100
Scarcella, et al. (2011)	Y	Y	Y	N	Y	Y	Y	Y	Y	89
Tesfay, et al. (2020).	Y	Y	Y	N	Y	Y	Y	Y	Y	89
van Oosterhout, et al.	Y	Y	Y	Y	Y	Y	Y	Y	Y	100

Q= Question Y=Yes; N= No; NA= Not applicable, UC=unclear

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Annex table 1 Improvements in nutritional status and recovery and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food type	Improvement in nutritional status or recovery	Factors that affect nutritional recovery
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUTF	328 (62.2%) recovered. **Measurement**: BMI>=18.5 kg/m^2 in two subsequent measurements	The factors that were associated with nutritional recovery were: - female (adjusted odds ratio (AOR) and 95% confidence interval (CI)= 2.16 (1.340-3.473)) - educated (AOR and 95%= 1.82(1.110-2.983)) - moderate undernutrition (AOR and 95%= 2.594 (1.251-5.377)) - mild undernutrition (AOR and 95% =14.25 (7.074-28.742)) - WHO stage I and II (AOR and 95% CI =3.345 (1.652-6.772)) - no opportunistic infection during admission (AOR and 95% =2.791(1.047-7.439))
Mallewa, et al 2018 31, Africa	Randomized clinical trial	-Adults -At least five years with HIV infection and CD4 cell count fewer than -100	1805 -897 to treatment -908 to non RUSF	Peri-urban regional hospital **Food type**: RUTF and RUSF	Adults and adolescents aged 13 years and older in the RUSF group had significantly greater gains in BMI, and MUAC than the no-RUSF group	Not assessed

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		-Starting ART at the health facilities			(p=0·004, and 0·03, respectively). **Measurement:** Improvement in weight, BMI and MUAC (no cut off point)	
Nagata et al 2014 40, Kenya	Retrospective cohort study	-non pregnant adult patients -BMI less than 18.5 kg/m^2	1017	Rural village **Food type**: fortified corn-soy flour supplements	On average, enrolled clients gained 2.01 kg in weight and 0.73 kg/m^2 in BMI over follow-up (mean 100 days) with the highest gain among those severely undernourished. - Only 13.1% nutritional recovery according to criteria **Measurement**: BMI.20 kg/m^2	The factors associated with BMI >20 Kg/m^2 were - Greater BMI at baseline 1.52 (1.23–1.86) - younger age, 0.97 (0.94–1.00) - male gender, (0.51 (0.30–0.86)) - not requiring highly active antiretroviral therapy (HAART) (4.18 (2.42–7.22)

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Ndekha et al 2009 34, Malawi	Randomised control trial	- Adults living with HIV - BMI<18.5 Kg/m^2	491	Clinic based **Food type**: fortified corn-soy flour supplements	After 14 weeks, patients receiving fortified spread had a greater increase in BMI and fat-free body mass than those receiving corn-soy blend: 2.2 (SD 1.9) v 1.7 (SD 1.6). Measurement: Gain in BMI (no cut off point)	No significant differences in the CD4 count, HIV viral load, assessment of quality of life, or adherence to antiretroviral therapy were noted between the two groups
Oguntibeju et al 2007 41, South Africa	Retrospective cohort study	- non pregnant adults with HIV, non-diabetic and non-hypertensive - not on ART - CD4 cell count 100-350 cells/mm^3	35	Not specified **Food type**: extract of hypoxis, grape fruit seed extract, sitosterol and sitosterolin, beta-carotene, vitamin E, vitamin, vitamin B1, vitamin B2, vitamin B12, nicotinamide, vitamin C, olive green	19 (67.9%) of the 28 patients had a BMI within the normal range after nutrient intervention **Measurement:** BMI>=18.5kg/m^2	- significant positive correlation between the BMI and fat percentage (statistics not reported) - CD4+T cell count showed no correlation with any of the anthropometric indices - high viral load has significant negative correlation with the lean body mass and BMI (statistics not reported)

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				leaf extract, folic acid and natural antioxidant (biocydin)		
Olsen et al 2014 33, Ethiopia	Randomised control trial	- Age greater than 18 years - BMI greater than 16 Kg/m^2	453	Referral hospital **Food type**: Plump sup	At three months, participants receiving the supplements containing whey or soy had increased their lean body mass by 0.85 kg (95% confidence interval 0.16 kg to 1.53 kg) and 0.97 kg (0.29 kg to 1.64 kg), respectively, more than controls. **Measurement**: Weight gain and BMI (no cut off point)	- No effect on physical activity - The effects of the two supplements, however, were not significantly different in direct comparison
Rawat et al, 2014 35, Uganda	Prospective cohort	- Food insecure people living with HIV who are new to ART. - employment status and income	904	Community based **Food type**: WFP Rations	Over 12 months, food assistance significantly increased body mass index by 0.6 kg/m2 (P, 0.01) and mid-upper arm	- no impact on CD4 count, Hb concentrations, or Individual Dietary Diversity Score - at the household level, food assistance increased the Household Food Insecurity Access Scale, by 2.1 points (P, 0.01) - food assistance increased BMI by 0.6 kg/m2 and MUAC by 6.7 mm.

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		- housing characteristics and past experience of food insecurity			circumference by 6.7 mm (P, 0.05). **Measurement:** Change in BMI (no cut off points)	- no statistically significant impact of food assistance on Hb concentration or CD4 count
Sadler, et al, 25 Ethiopia	Comparative follow up study. Report, not peer reviewed	Adults living with HIV - BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 interventio n were included in the final analysis	Health facilities / health centres **Food type**: RUTF/RUSF	636 (32.6) graduated in the intervention group and 120 (18.8) in the comparison group **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Factors that were associated with nutritional recovery were: - being in the intervention group 2.4 (1.6-3.4) - female 1.4 (1.0-1.9) - attending in a hospital 0.5 (0.4-0.7)
Scarcella. et al 43, Mozambique	Randomised control trial	- Adult HIV patients - BMI <18.5 Kg/m^2 - presence of wasting - TB co-infection - insufficient access to food - CD4 count < 200	106	Public Health Centre **Food type**: - food package consisted of a ration of cereals (rice or corn) of around 15kg; pulses (5kg); peanuts (3kg); sugar (2kg); sunflower seed oil (3Lt.);	In the cohort not in program the mean BMI values showed an increase but not significant (initial value: 21.9±2.9; final value: 22.5±3.3) **Measurement**: Increase in BMI (no cut off point)	- haemoglobin showed an increase but not significant - in the intervention group both, BMI and Haemoglobin shown a significant increase - increase in BMI was statistically associated with the dietary diversity score in the intervention group

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				at least 7kg of fortified corn soy blend		
Tesfay et al, 42 Ethiopia	Retrospective cohort study	- Adults living with HIV - BMI<18.5	1757	Hospitals **Food type**: RUTF/RUSF	55.3% (95% CI = 53.2–57.4) recovered from undernutrition **Measurement:** BMI>=18.5 kg/m^2 in two subsequent measurements	Not assessed
van Oosterhout et al 2010 37, Malawi	Retrospective cohort study	-Adults aged 18 years and older -BMI 18.5 kg/m^2 -ART naïve or - WHO clinical stage III or IV Or - CD4 count B250 cells/ml irrespective of clinical stage	593	Centeral hospital **Food type**: corn/soy blended flour (CSB) or ready-to-use fortified spread (RUFS), a peanut butter-based product	- Linear regression analysis showed that increase in BMI was greatest in the RUSF group and better in the CSB group than in those receiving no food supplementation at 14 weeks. - These differences were no longer significant at 26 weeks **Measurement:** Increase in BMI	- no longer a significant effect of supplementary feeding on an increase in BMI, 12 weeks after the feeding was stopped. - factors that were associated with nutritional recovery or improved BMI was not assessed.

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Annex table 2: Shows program failure (default + nonresponse +death) after enrolment in a nutritional program in sub Saharan Africa

Author, year and country	Study design	Study population	N	Setting and type of food	Nutritional program failure	Factors that affect nutritional program failure
Ahou et al 2011 38, Kenya and uganda	Retrospective cohort study	Adults aged 15 years and older living with HIV BMI less than 17 kg/m^2 and bilateral pitting oedema	1106	Regional hospital Type of food: RUTF/RUSF	An overall total of 531 of (48.0%) patients failed nutrition therapy	Factors that were associated with program failure were: - male gender (OR = 1.5, 95% CI 1.2-2.0), - patients with severe undernutrition at nutrition programme enrolment (OR = 2.2, 95% CI 1.7-2.8), and - those never started on ART (OR = 4.5, 95% CI 2.7-7.7) for those eligible; OR = 1.6, 95% CI 1.0-2.5 for those ineligible for ART at enrolment
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital Type of food: RUTF/RUSF	Overall program failure rate was 196 (37.6%)	The factors that were associated with nutritional program failure were not assessed.

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Annex table 3: Nonresponse to the nutritional program and its determinants in sub-Saharan Africa

Author, year and country	Study design	Study population	N	Setting and food	Nonresponse to the nutritional program	Factors that contribute to non-response to the nutritional program
Berhe, et al 2014 44, Ethiopia	Retrospective cohort study	People living with HIV and BMI less than 18.5 kg/m^2	531	Regional hospital **Food type**: RUFT/RUSF	- 155 (29.6%) non-respondent - **Measurement**: not reaching 18.5 kg/m^2	The factors that affect non-response were not assessed
Sadler, et al , 25 Ethiopia	Comparative follow up study. not peer reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each. 1956 and 639 were including in the final analysis	Health facilities / health centres **Food type**: RUFT/RUSF	- 1316 (67.4%) non-respondent in the intervention group and 519 (81.2) become non-respondent in the comparison group - **Measurement**: not reaching 18.5 kg/m^2	Factors that were associated with nutritional non-response were: - attending the program at a hospital was associated with an increased likelihood of non-response (OR=1.75, p=0.021), - severe food insecurity was associated with a decreased likelihood of non-response (OR=0.46, p=0.016) compared to those who were "food secure.
Tesfay et al , 42 Ethiopia	Retrospective cohort study	Adults living with HIV and-BMI<18.5	1757	Hospitals **Food type**: RUFT/RUSF	- 21% (95% CI = 19.7–23.4) completed the program but failed to recover from undernutrition. - **Measurement**: not reaching 18.5 kg/m^2	- urban residence, (AOR = 1.46, 95% CI = 1.4–2.91), - attending Shul (AOR = 2.92, 95% CI = 2.04–4.19) and - Lemlem Karl (AOR = 1.49, 95% CI 1.05–2.11) hospitals

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Annex table 4: Default from the nutritional program and its determinants in sub-Saharan Africa

Author, year, and country	Study design	Study population	N	Setting	Default to the nutritional program	Factors that contribute to default to the nutritional program
Sadler, 25 et al , Ethiopia	Comparative follows up study. not peer-reviewed	Adults living with HIV and BMI<18.5kg/m^2	2059 each group but 1956 from the control and 639 intervention were included in the final analysis	Health facilities / health centres	70.6% defaulted from the nutritional program among the intervention group	- severe acute undernutrition (OR: 1.89, P-value=0.003)
Tesfay et al 42 , Ethiopia	Retrospective cohort study	Adults living with HIV and BMI<18.5	1757	Hospitals	19% (95% CI, 17.3–20.7) did not complete the program.	Factors associated with program incompleteness. - living in urban areas (AOR = 1.44, 95% CI = 1.05– 1.97), - employed (AOR = 1.39, 95% CI = 1.01–1.93), - attending Shul (AOR = 4.6, 95% CI = 3.15–6.71) and Lemlem Karl (AOR = 2.5, 95% CI = 1.69–3.71) hospitals, - in clinical stages II (AOR = 2.49, 95% CI = 1.59–3.91) and III (AOR = 1.46(1.02–2.07), - on ART for less than six months (AOR = 1.61, 95% CI = 1.09–2.39), - anaemic (AOR = 1.77, 95% = 1.29–2.41), and - severe acute undernutrition at enrolment (AOR = 6.43, 95% CI = 4.69–8.3)

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3-4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	6
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	7
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6-7
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	7
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	N/A

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Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	8
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	N/A
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	10
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	9
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	26
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	30
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	30
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	No funding

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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